7

Momentum

In this chapter, we will learn about

- different types of collisions
- the meaning of momentum
- the law of conservation of momentum
- the relationship between net force and the change in momentum of an object

7.1 Conservation of momentum

Flipped classroom
Watch this and answer the questions.

Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Colliding marbles

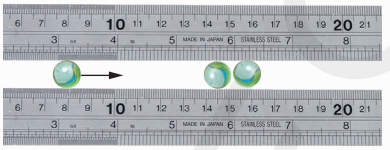
What you need

2 rulers

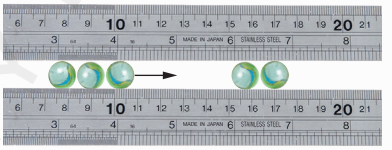
5 marbles (same size and mass)

What to do

1 Make a narrow passage using 2 rulers. Push a marble towards 2 stationary marbles placed together.



2 Repeat with different numbers of marbles, e.g. pushing 3 marbles towards 2 stationary marbles.



What you find

How many marbles move after each collision?

In the above activity, suppose each marble is 0.3 kg. When one marble of mass 0.3 kg moving at velocity 4 m s^{-1} to the right collides with two stationary marbles, there could be many outcomes that satisfy the law of conservation of energy (Fig 7.1a).

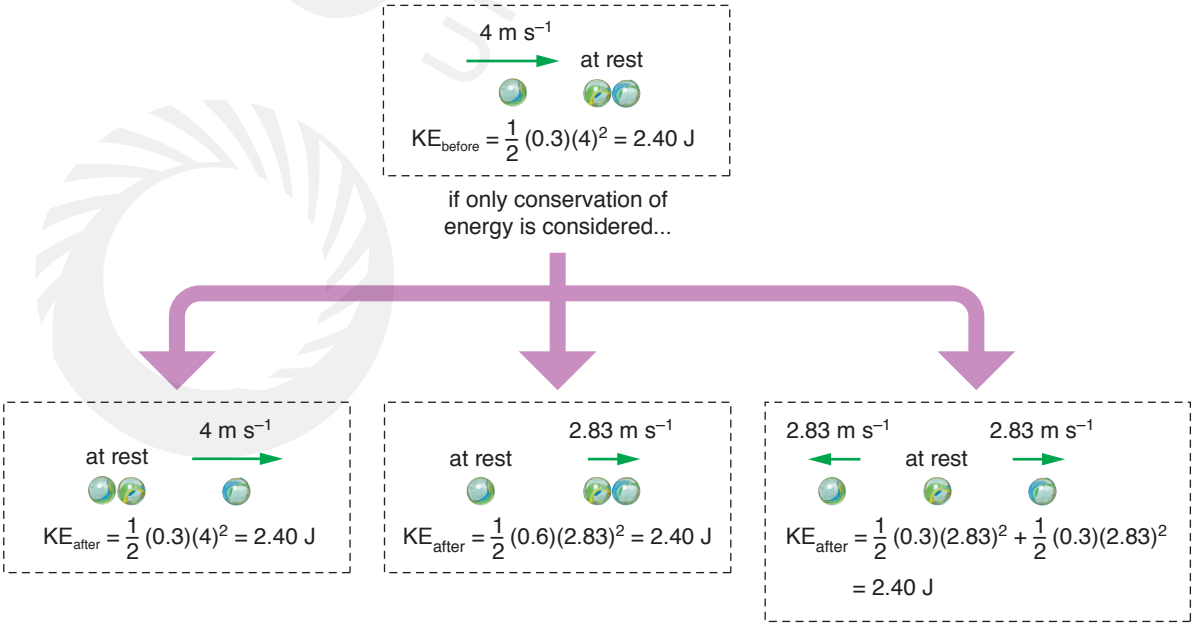


Fig 7.1a *Some outcomes that satisfy the law of conservation of energy, assuming m = 0.3 kg and u = 4 m s^{-1}.*

However, the activity shows that the number of marbles that move after a **collision** is always equal to those move before. This cannot be explained by the law of conservation of energy alone. There seems to be another law that governs the collisions of objects (Fig 7.1b). In this section, we shall find this law.

Fig 7.1b *Examples of collisions.*

DSE goal

Realize momentum as a quantity of motion of an object and define momentum $p = mv$.

DSE exam

23(1A)Q4

1 Momentum

It turned out that the quantity mass × velocity, i.e. mv, plays an important role in studying collisions. This quantity is called **momentum**. It is usually represented by the letter p. Its SI unit is kilogram metre per second (kg m s^{-1}).

$$p = mv$$

momentum (kg m s^{-1}) · mass (kg) · velocity (m s^{-1})

According to the definition, an object has a greater momentum if it has a greater velocity (Fig 7.1c).

(i) A slow-moving ball. *(ii) A fast-moving ball.*

Fig 7.1c *The fast-moving ball has a greater momentum.*

collision 碰撞 momentum (pl. momenta) 動量

For two objects moving at the same velocity, the one with a larger mass has a greater momentum (Fig 7.1d).

Fig 7.1d *The more massive elephant has a greater momentum than the less massive one when they run at the same velocity.*

Note that:

1 Momentum is a vector. It has the same direction as the velocity.

2 For objects moving along a straight line, the directions of their momenta can be represented by positive and negative signs. We can calculate the total momentum of the objects by adding their momenta (Fig 7.1e).

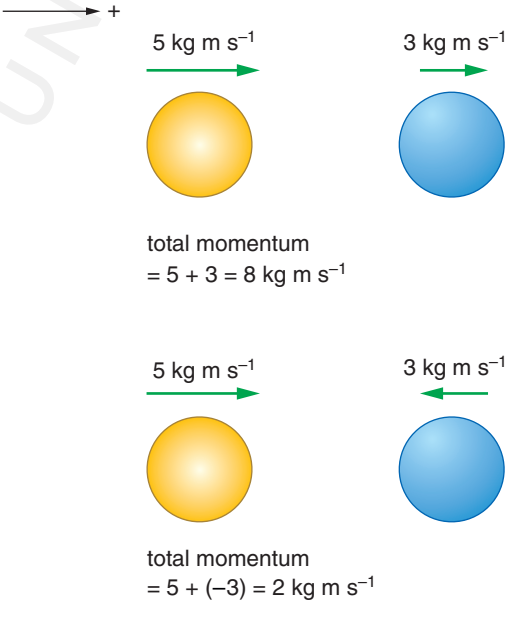


Fig 7.1e *Calculating total momentum.*

Q38 Exam Report: This question tested candidates' knowledge and understanding in Mechanics in the context of bungee jumping. The overall performance was only fair.

(a) (i) Most candidates knew that the jumper was in free fall at the start. However, only the most able ones described correctly the change in acceleration after the cord began to stretch. Many candidates wrongly thought that the jumper began to decelerate once the cord stretched.

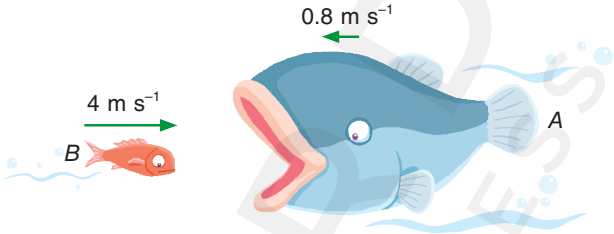
(ii) Most demonstrated an understanding that gravitational potential energy changed to kinetic energy during the fall. However, not many mentioned that all the gravitational potential energy changed to elastic potential energy in the cord at the lowest point.

(b) Only the more able candidates explained clearly that the elastic cord lengthened the stopping time, hence reducing the net force acting on the jumper.

(c) Candidates knew that contact area was larger for a 'full body harness' but few were able to apply the concept of pressure in their explanations.

Example 1 Calculating momenta

Fishes A and B swim towards each other. A of mass 5 kg swims at 0.8 m s^{-1} towards the left. B of mass 0.6 kg swims at 4 m s^{-1} towards the right. Find their individual momenta and total momentum.



Solution

Take the direction towards the left as positive.

Apply $p = mv$.

$p_A = 5 \times 0.8 = 4$ kg m s^{-1}

The momentum of A is 4 kg m s^{-1} (leftwards).

$p_B = 0.6 \times (-4) = -2.4$ kg m s^{-1}

The momentum of B is 2.4 kg m s^{-1} (rightwards).

$p_A + p_B = 4 + (-2.4) = 1.6$ kg m s^{-1}

Their total momentum is 2.4 kg m s^{-1} (leftwards).

Quick check 1

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 The direction of momentum of an object must be the same as

A the displacement of the object.

B the velocity of the object.

C the acceleration of the object.

D the net force acting on the object.

2 Several objects each of mass m move at velocity v and have a momentum p. Complete Table a.

Object	m	v	p
A	0.5 kg	6 m s^{-1}	
B		2.5 m s^{-1}	8 kg m s^{-1}
C	0.02 kg		–3 kg m s^{-1}

Table a

3 A man and a bird are shown in Figure a.

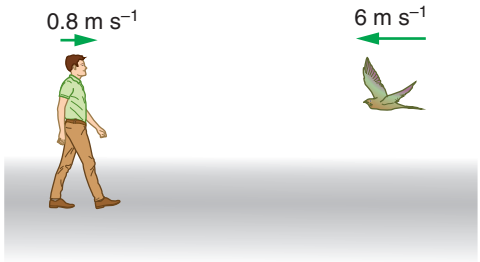


Fig a

Taking the direction towards the right as positive, complete Table b.

Object	m	v	p
man	60 kg		
bird	100 g		

Table b

DSE goal

State the law of conservation of momentum.

Solve problems involving momentum in one dimension.

DSE exam

17(1B)Q2, 18(1A)Q12, 20(1A)Q8, 23(1A)Q11

Keep a safe distance from the trolleys.

Simulation Video

→ This simulation simulates Expt 7a.

→ This video demonstrates Expt 7a.

2 Law of conservation of momentum

In the following experiments, we shall use trolleys as colliding objects to study their momenta before and after collisions.

Experiment 7a / Colliding trolleys (with Velcro attachments)

1 Set up the apparatus as shown, with the Velcro attachments on the trolleys facing each other (Fig a).

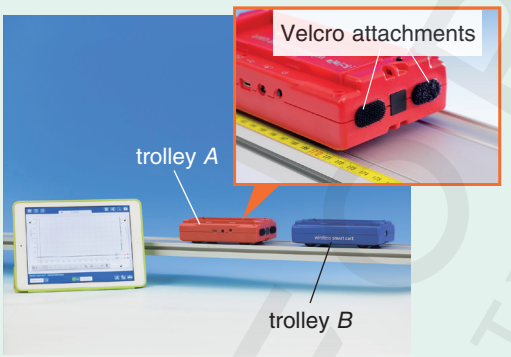


Fig a

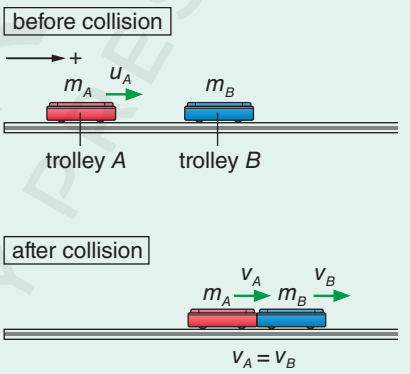


Fig b

2 Start data-logging. Push trolley *A* sharply so that it collides with stationary trolley *B*. The trolleys stick together after the collision (Fig b).

3 Record the velocities of the trolleys just before and after the collision.

4 Change the mass of trolley *A* and repeat the experiment.

Precaution The precautions aim to minimize the external forces.

Lubricate the wheels of the trolleys to minimize friction. Level the track.

Discussion

Which quantity is conserved in the collisions?

Some sample results of the experiment are shown in Figure 7.1f and Table 7.1a. Explain to Ss that they should read the velocity just before and just after the collision from the *v*–*t* graph.

Fig 7.1f *A sample v–t graph in Experiment 7a (m_A = 1 kg, m_B = 0.5 kg).*

Trial	m_A / kg	m_B / kg	Before collision ($u_B = 0$)	After collision ($v_B = v_A$)
			u_A / m s^{-1}	v_A / m s^{-1}
1	0.50	0.50	0.12	0.06
2	1.00	0.50	0.36	0.24
3	1.50	0.50	0.40	0.29

Table 7.1a *Sample results of Experiment 7a (trial 2 corresponds to Fig 7.1f).*

Table 7.1b shows the analysis of the results.

Trial	p before collision / kg m s^{-1}			p after collision / kg m s^{-1}		
	p_A	p_B	Total p	p_A	p_B	Total p
1	0.06	0	0.06	0.03	0.03	0.06
2	0.36	0	0.36	0.24	0.12	0.36
3	0.60	0	0.60	0.44	0.15	0.59

After allowing for experimental errors, the total momentum before collision is the same as the total momentum after collision.

Table 7.1b *Analysis of the results of Experiment 7a.*

The momentum of each trolley changes after the collision. But the results, after allowing for experimental errors, show that the total momentum of the two trolleys is conserved. Will the results be the same in other types of collisions?

A possible source of experimental error is the resistive force acting on the trolleys during the collision.

Keep a safe distance from the trolleys.

Video

→ This video demonstrates Expt 7b.

Experiment 7b Colliding trolleys (with rubber bumper)

1 Set up the apparatus as shown, with a rubber bumper attached to trolley *A* (Fig a).

2 Take the direction towards the right as positive when taking measurements.

3 Start data-logging. Push trolley *A* towards stationary trolley *B* (Fig b).

4 Record the velocities of the trolleys just before and after the collision.

5 Change the mass of trolley *B* and repeat the experiment.

Fig a

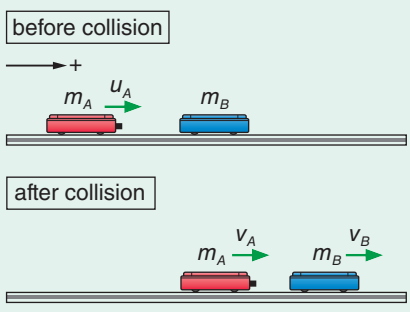


Fig b

Precaution

Lubricate the wheels of the trolleys to minimize friction. Level the track.

Discussion

Is the total momentum of the trolleys conserved in the collisions?

For an elastic collision, if $m_A = m_B$, trolley *A* will come to a stop after colliding with stationary trolley *B*. However, slight variation in the trolley mass and possible imperfect levelling of the runway may not allow such an observation.

Keep a safe distance from the trolleys.

Simulation Video

→ This simulation simulates Expt 7c.

→ This video demonstrates Expt 7c.

Experiment 7c / Colliding trolleys (with magnetic bumpers)

1 Attach magnetic bumpers to the trolleys with the bumpers facing each other (Fig a). The trolleys repel each other when they come close.

2 Repeat steps 2–4 in Experiment 7b.

3 Change the mass of trolley *A* and repeat the experiment.

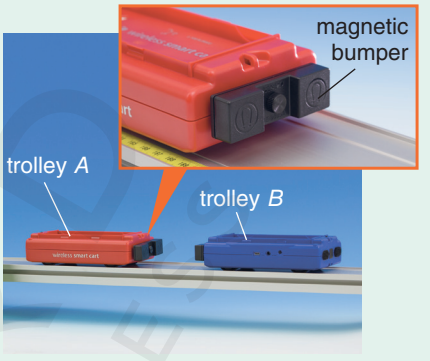


Fig a

Precaution

Lubricate the wheels of the trolleys to minimize friction. Level the track.

Discussion

Is the total momentum of the trolleys conserved in the collisions?

A sample of the *v*–*t* graphs obtained in the above two experiments is shown below (Fig 7.1g and 7.1h). Tables 7.1c and 7.1d show the analysis of the experiments.

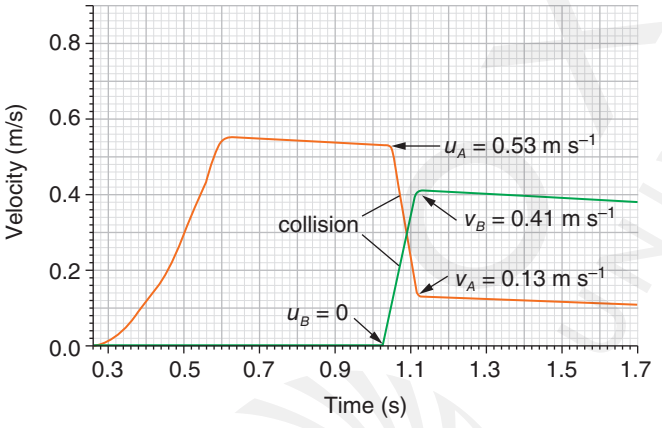


Fig 7.1g *A sample v–t graph in Experiment 7b ($m_A = m_B = 0.5$ kg).*

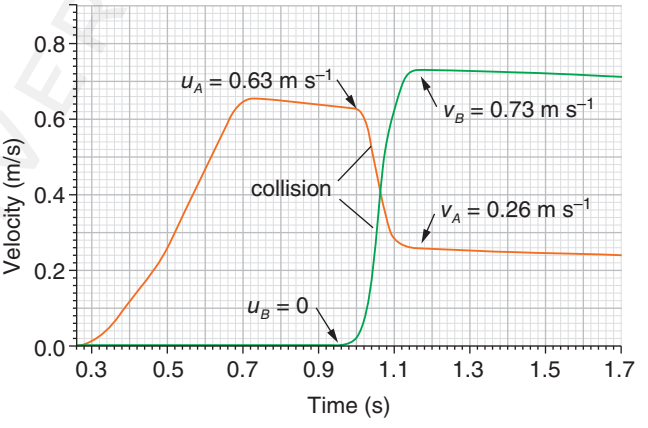


Fig 7.1h *A sample v–t graph in Experiment 7c (m_A = 1 kg, m_B = 0.5 kg).*

Trial	m_A / kg	m_B / kg	Before collision			After collision		
			u_A / m s^{-1}	u_B / m s^{-1}	Total p / kg m s^{-1}	v_A / m s^{-1}	v_B / m s^{-1}	Total p / kg m s^{-1}
1	0.50	0.50	0.53	0	0.27	0.13	0.41	0.27
2	0.50	1.00	0.64	0	0.32	0.10	0.27	0.32
3	0.50	1.50	0.44	0	0.22	0.04	0.13	0.22

After allowing for experimental errors, the total momentum before collision is the same as the total momentum after collision.

Table 7.1c *Analysis of the results of Experiment 7b (trial 1 corresponds to Fig 7.1g).*

Trial	m_A / kg	m_B / kg	Before collision			After collision		
			u_A / m s^{-1}	u_B / m s^{-1}	Total p / kg m s^{-1}	v_A / m s^{-1}	v_B / m s^{-1}	Total p / kg m s^{-1}
1	0.50	0.50	0.29	0	0.15	0	0.29	0.15
2	1.00	0.50	0.63	0	0.63	0.26	0.73	0.63
3	1.50	0.50	0.53	0	0.80	0.30	0.68	0.79

After allowing for experimental errors, the total momentum before collision is the same as the total momentum after collision.

Table 7.1d *Analysis of the results of Experiment 7c (trial 2 corresponds to Fig 7.1h).*

Again, after allowing for experimental errors, the results show that the total momentum of the trolleys is conserved in both Experiments 7b and 7c.

Let us study one more case: explosion. It can be considered as a special kind of collision, in which an object explodes into two or more fragments.

Keep a safe distance from the trolleys.

Simulation Video

→ This simulation simulates Expt 7d.

→ This video demonstrates Expt 7d.

Experiment 7d 'Explosion'

1 Set up the apparatus as shown (Fig a). One of the trolleys is equipped with a spring-loaded plunger.

2 Take the direction towards the right as positive when taking measurements.

3 Start data-logging. Gently hit the plunger to release it so that the two trolleys 'explode' and move apart (Fig b).

4 Record the velocities of the trolleys just after the explosion.

5 Change the mass of trolley A and repeat the experiment.

Fig a

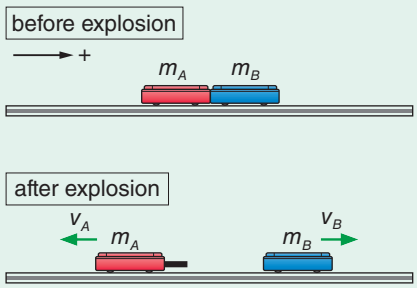


Fig b

Precaution

Lubricate the wheels of the trolleys to minimize friction. Level the track.

Discussion

Is the total momentum of the trolleys conserved in the explosions?

Some sample results in the experiment are shown in Figure 7.1i and Table 7.1e on p.286.

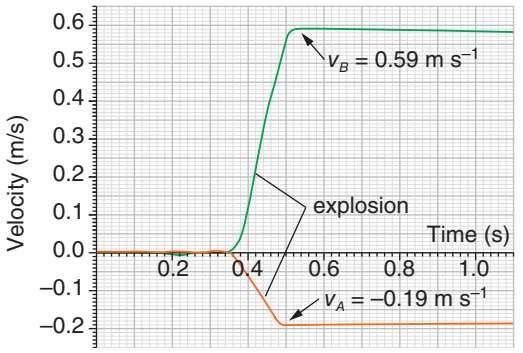


Fig 7.1i *A sample v–t graph in Experiment 7d (m_A = 1.5 kg, m_B = 0.5 kg).*

Trial	m_A / kg	m_B / kg	After explosion		
			v_A / m s^{-1}	v_B / m s^{-1}	Total p / kg m s^{-1}
1	0.50	0.50	–0.54	0.58	0.02
2	1.00	0.50	–0.30	0.63	0.02
3	1.50	0.50	–0.19	0.59	0.01

Table 7.1e *Analysis of the results of Experiment 7d. Momenta before explosion are all zero (trial 3 corresponds to Fig 7.1i).*

After explosion, although the momentum of each trolley is large, their total momentum is almost zero. The results, after allowing for experimental errors, show that the total momentum of the two trolleys is conserved in an explosion.

By considering the colliding objects as a system, the above results suggest that the total momentum of a system is conserved in all types of collisions. This is the **law of conservation of momentum**:

In fact, the total momentum of a system will change if it is acted on by an external net force. This will be discussed in Chapter 7.2.

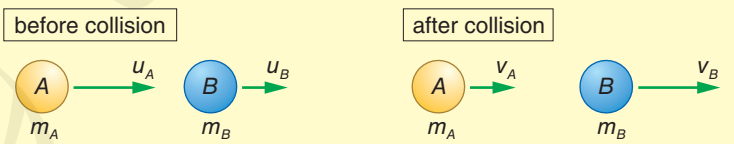
The total momentum of a system is conserved, provided that there is no external net force acting on the system.

Colliding head-on: objects moving along a straight line before and after the collision

For a system consisting of two objects A and B which collide head-on, the law can be expressed as

total momentum before collision = total momentum after collision

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$



Cross-link
You learned about system, internal force and external force in Chapter 4.1.

Note the following (Fig 7.1j):

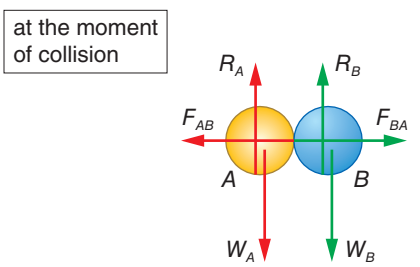


Fig 7.1j *Forces acting on A and B.*

law of conservation of momentum 動量守恆定律

The forces F_{AB} and F_{BA} acting between the objects are internal forces.

1 If we consider a **system that consists of *A* and *B***, the external forces to the system include the weight of A and B and the normal force from the floor. If the weight and the normal force cancel each other out, the external net force is zero and therefore the total momentum of the objects is conserved.

2 If we consider a **system that consists of *A*** only (Fig 7.1k), F_{AB} will become the external force. The external net force is not zero and therefore the momentum of A does change.

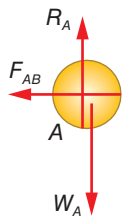


Fig 7.1k *Forces acting on A.*

The law of conservation of momentum is very useful in solving collision problems. Newton's second law of motion is usually not suitable to be applied in such problems as the collision forces between two colliding objects are usually not constant. See the following examples.

→ Teaching notes and exercises are available on OUP web.

Skill *Problem-solving*

Applying conservation of momentum

① Define the positive direction.

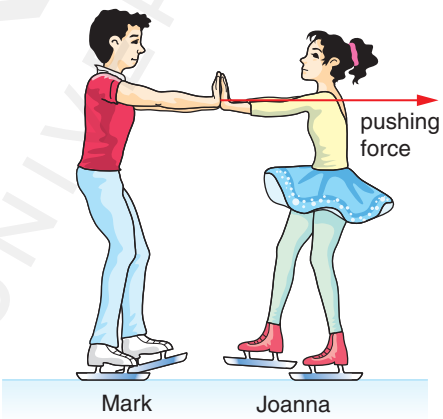
② Check the external forces acting on the system. Conservation of momentum is valid only if the external net force is zero.

③ Set an equation with the conservation of momentum.

④ Solve the equation.

Example 2 Skaters pushing each other

Mark and Joanna are standing on ice. To get them moving, Mark pushes Joanna with a constant force for 0.5 s. Joanna moves at 0.9 m s^{-1} towards the right after the push. Mark's mass is 60 kg and Joanna's mass is 55 kg. Assume that the friction acting on them by the surface of the ice is negligible. What is Mark's velocity after pushing Joanna?



Solution

Take the direction towards the right as positive.

By the law of conservation of momentum,

Tip
$u_M = 0$ and $u_J = 0$

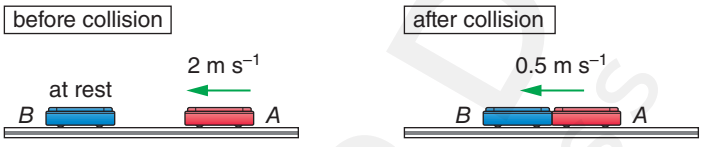
$$m_M u_M + m_J u_J = m_M v_M + m_J v_J$$
$$0 + 0 = 60 v_M + 55 \times 0.9$$
$$v_M = -0.825 \text{ m s}^{-1}$$

Try to work out the solution using Newton's second law of motion.

Mark's velocity is 0.825 m s^{-1} (leftwards) after pushing Joanna.

Example 3 Trolleys sticking together after collision

Trolley A of mass 0.5 kg moves at 2 m s^{-1} towards the left on a smooth horizontal track. After colliding with stationary trolley B, they move together at 0.4 m s^{-1} towards the left.



(a) Find the mass m of trolley B.

(b) Find the total KE of the trolleys before and after the collision.

Solution

(a) Take the direction towards the left as positive.

By the law of conservation of momentum,

$m_A u_A + m_B u_B = (m_A + m_B)v$

$0.5 \times 2 + 0 = (0.5 + m) \times 0.4$

$m = 2$ kg

(b) Total KE before collision $= 0 + \frac{1}{2} \times 0.5 \times 2^2 = 0.5$ J

Total KE after collision $= \frac{1}{2} \times (0.5 + 0.5) \times 0.5^2 = 0.125$ J

Tip
Since the track is horizontal and smooth, the external net force acting on the system of colliding trolleys is zero.

Tip
The two trolleys moving together after the collision can be considered as a single object.

Learn by practice 1

1 Refer to Figure a.

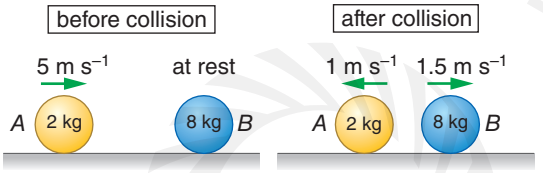


Fig a

Taking the direction towards the right as positive, find the following quantities.

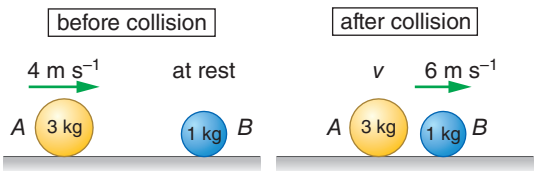
$m_A =$ _____ $u_A =$ _____ $v_A =$ _____

$m_B =$ _____ $u_B =$ _____ $v_B =$ _____

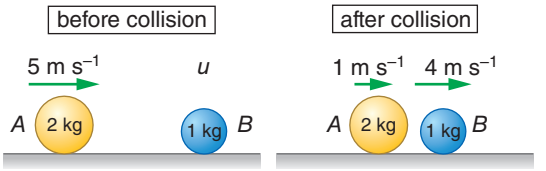
2 Nancy of mass 50 kg holds a 2-kg ball and skates forwards at 3 m s^{-1}. Then she throws the ball so that it moves forwards at 10 m s^{-1}. What is her velocity after throwing the ball?

3 In each case, find the **unknown velocity**. Also find the **total KE of the balls before and after collision**. Assume no external net force acts on the system of the two balls.

(a)



(b)



DSE goal

Distinguish between elastic and inelastic collision.

DSE exam

13(1A)Q10, 14(1A)Q7, 16(1A)Q12, 19(1A)Q11, 22(1A)Q12

3 Conservation of momentum and conservation of energy

From Example 3 on p.288, you may notice that the total KE of colliding objects may not be conserved. By considering the total KE of the colliding objects before and after a collision, we can categorize collisions into two main types.

If the total KE of the colliding objects is conserved, the collision is an elastic collision. Otherwise, the collision is an inelastic collision.

The total KE of the objects is not conserved. ▶ If the colliding objects stick together after a collision, we say that the collision is a *completely / perfectly inelastic collision*.

For example, if a ball falls and bounces back to the same height, the collision is elastic. If an egg is dropped to the ground, it sticks to the ground and does not rebound at all. This collision is completely inelastic (Fig 7.1l).

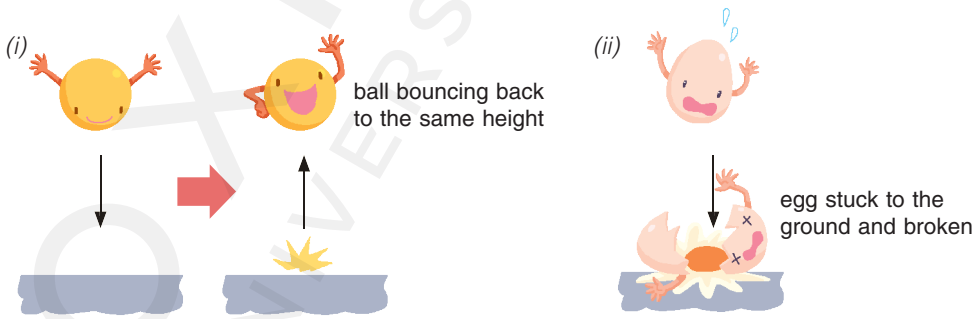


Fig 7.1l *(i) Elastic collision and (ii) completely inelastic collision.*

Similar energy conversion happens in Experiment 7d on p.285 when the plunger is released, and in Example 2 on p.287 when the two skaters push one another. ▶ In explosion, some other forms of energy is converted to the KE of the objects so that the objects can move apart. The total KE of the objects is not conserved in the process.

Table 7.1f summarizes whether the total momentum p and the total KE are conserved in different types of collisions.

→ This simulation allows Ss to freely explore different types of interaction between two trolleys. It is available on OUP web in PC version (.exe) only.

Simulation

Experiment	Type of collision	Total p conserved?	Total KE conserved?
7a	completely inelastic collision	✓	✗
7b	inelastic collision	✓	✗
7c	elastic collision	✓	✓
7d	explosion	✓	✗

In completely inelastic collision, the loss in KE is maximum. ▶

The total KE increases because the elastic PE stored in the spring of the plunger is converted into the KE of the trolleys. ▶

Table 7.1f *Total p and total KE in different types of collisions.*

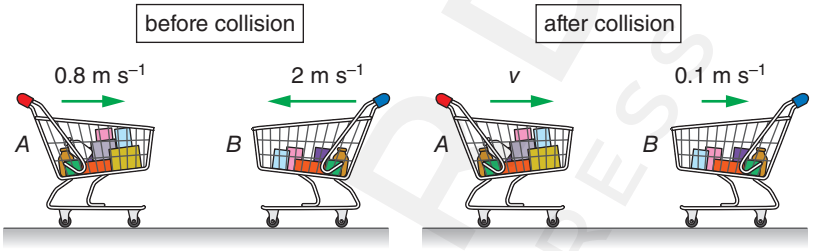
elastic collision 彈性碰撞　　inelastic collision 非彈性碰撞
completely / perfectly inelastic collision 完全非彈性碰撞

Simulation

→ This simulation allows Ss to investigate if momentum is conserved when a mass is added to a moving trolley or when a trolley sheds a mass while moving.

Example 4 Trolleys moving in opposite directions

Trolley A of mass 10 kg moves at 0.8 m s^{-1} towards the right. Trolley B of mass 6 kg moves at 2 m s^{-1} towards the left. After colliding head-on, B moves at 0.1 m s^{-1} towards the right. Assume the friction acting on the trolleys is negligible.



(a) What is A's velocity after the collision?

(b) What is the loss in the total KE of the trolleys? What kind of collision is this?

(c) If the collision is completely inelastic, what is the loss in the total KE of the trolleys?

Solution

Take the direction towards the right as positive.

(a) By the law of conservation of momentum,

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$10 \times 0.8 + 6 \times (-2) = 10 \times v_A + 6 \times 0.1$$

$$v_A = -0.46 \text{ m s}^{-1}$$

A's velocity is 0.46 m s^{-1} (towards the left).

(b) Total KE before collision $= \frac{1}{2} \times 10 \times 0.8^2 + \frac{1}{2} \times 6 \times 2^2 = 15.2$ J

Total KE after collision $= \frac{1}{2} \times 10 \times 0.46^2 + \frac{1}{2} \times 6 \times 0.1^2 = 1.09$ J

Loss in total KE = 15.2 − 1.09 = 14.1 J

This is an inelastic collision.

(c) By the law of conservation of momentum,

$$m_A u_A + m_B u_B = (m_A + m_B) v_B$$

$$10 \times 0.8 + 6 \times (-2) = (10 + 6) \times v$$

$$v = -0.25 \text{ m s}^{-1}$$

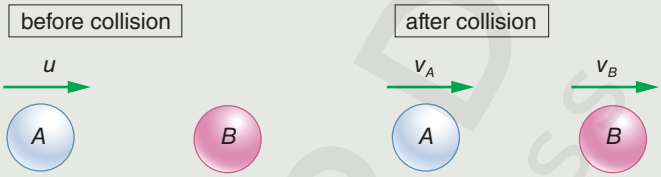
Total KE after collision $= \frac{1}{2} \times (10 + 6) \times 0.25^2 = 0.5$ J

Loss in total KE = 15.2 − 0.5 = 14.7 J

Know more

Special cases of elastic collision

Suppose object A of mass m moves forwards at speed u and collides head-on with stationary object B of mass M. The collision is elastic. In the special case when $m = M$, A will stop and B will move forwards at speed u. This can be proved as follows:



By the law of conservation of momentum,

$mu + 0 = mv_A + mv_B$

$u = v_A + v_B$ ..... (1)

Since the total KE is conserved in an elastic collision,

$\frac{1}{2}mu^2 + 0 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2$

$u^2 = v_A^2 + v_B^2$ ..... (2)

Substitute (1) into (2):

$(v_A + v_B)^2 = v_A^2 + v_B^2$

$2v_Av_B = 0$

$v_A = 0$ or $v_B = 0$ (rejected as $v_A = u$ is not possible)

From (1),

$u = 0 + v_B$

$v_B = u$

Video

→ This video shows a stop shot.

The stop shot in a pool or a snooker game is an example of such collision.

If a stop shot is made, the white ball will collide head-on with the ball with number 10 and stop after the collision.

Two other special cases for elastic collision between objects A and B are worth noticing.

1 If $m \ll M$, A will rebound with speed u and B will remain at rest.

2 If $m \gg M$, A will continue to move forwards at speed u and B will move forwards at speed $2u$.

You may try to prove these by yourself.

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Identify the type of collision (elastic, inelastic or completely inelastic) in the following cases.

(a) A ball hits a wall and rebounds with the same speed.

(b) A car crashes into a lamp-post and stops.

(c) A ball falls to the ground from a height of 1 m and rebounds to a maximum height of 0.8 m.

Learn by practice 2

1 Object X has a mass of 2 kg and object Y has a mass of 4 kg. Their velocities before and after each collision between them are shown in the table (taking the direction towards the right as positive). No external net force acts on the system of the two objects. Complete the table.

Collision	Before collision		After collision	
	u_X / m s^{-1}	u_Y / m s^{-1}	v_X / m s^{-1}	v_Y / m s^{-1}
A	2		−2	−2
B	−4	0	−1	
C		1	1	−1.5

2 Determine the type of collision for each collision in Q1.

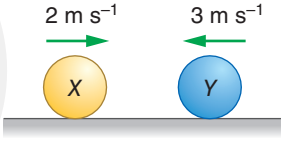
☆ HKDSE 2019 Paper 1A Q11

→ This video demonstrates how this question is solved.

Video

DSE practice 1 Colliding balls

On a smooth horizontal plane, balls X and Y collide head-on. X of mass 50 g moves at 2 m s^{-1} towards the right and Y of mass 200 g moves at 3 m s^{-1} towards the left just before collision. Which of the following is/are the possible result(s) after collision?



(1) X stops.

(2) Y stops.

(3) Y moves at 2 m s^{-1} towards the left.

A (1) only B (3) only

C (2) and (3) only D (1), (2) and (3)

Solution

Take the direction towards the right as positive.

Total p before collision

$= m_X u_X + m_Y u_Y$

$= 0.05 \times 2 + 0.2 \times (-3)$

$= -0.5$ kg m s^{-1}

Total KE before collision

$= \frac{1}{2} m_X u_X^2 + \frac{1}{2} m_Y u_Y^2$

$= \frac{1}{2} \times 0.05 \times 2^2 + \frac{1}{2} \times 0.2 \times 3^2$

$= 1$ J

For (1):

If X stops after the collision, Y must move towards the left so that total p is conserved. This means that Y passes through X, which is not possible.

Common mistake
Students may not be aware of this condition.

$\therefore$ (1) is not possible.

For (2):

By the law of conservation of momentum,

$m_X u_X + m_Y u_Y = m_X v_X + m_Y v_Y$

$-0.5 = 0.05 v_X + 0$

$v_X = -10$ m s^{-1}

Total KE after collision

$= \frac{1}{2} \times 0.05 \times 10^2 = 2.5$ J > total KE before collision

This violates the law of conservation of energy.

Common mistake
Students may not notice that the total KE after collision should not be larger than that before collision.

$\therefore$ (2) is not possible.

For (3):

By the law of conservation of momentum,

$m_X u_X + m_Y u_Y = m_X v_X + m_Y v_Y$

$-0.5 = 0.05 v_X + 0.2 \times (-2)$

$v_X = -2$ m s^{-1}

As v_X is the same as v_Y, Y will not pass through X.

Total KE after collision

$= \frac{1}{2} \times 0.05 \times 2^2 + \frac{1}{2} \times 0.2 \times 2^2 = 0.5$ J < total KE before collision

$\therefore$ (3) is possible.

$\therefore$ The answer is B.

▶ Revision 7 Q22 (p.322)

Right and wrong

Aaron stands on ice. He starts moving backwards after catching a ball thrown by Molly.

What would happen if Aaron does not catch the ball but blocks it with his fists so that it rebounds backwards?

✗ He will move backwards at a lower speed.

✓ He will move backwards at a higher speed.

By the law of conservation of momentum,

$m_A v_A + m_B v_B$ = total momentum before collision

Take the direction to the right as positive. If the ball rebounds, the momentum of the ball ($m_B v_B$) will become negative, and hence the momentum of Aaron ($m_A v_A$) will become larger, meaning that Aaron's velocity (v_A) will become larger.

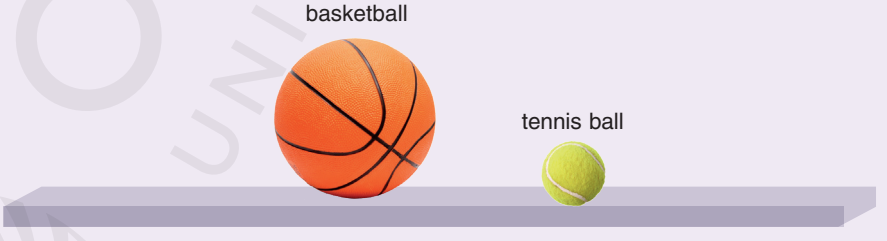
Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Bouncing balls

What you need



Beware of the bouncing balls.

What to do

Go to an open area. Put the tennis ball on top of the basketball and then drop the balls at the same time.

What you find

Do the balls return to the position that they are dropped from?

Try Revision 7 Q32 (p.325) to estimate how high the tennis ball reaches. ▶

4 Examples of conservation of momentum

a Recoil of guns and cannons

Due to the conservation of momentum, when a bullet is fired forwards, the gun *recoils* (moves backwards) so that the total momentum of the gun and the bullet remains zero (Fig 7.1m). The person holding the gun has to provide a force to stop the gun from moving backwards.

Similarly, when a cannonball is fired, the cannon recoils. Old cannons have massive bases to reduce the recoil velocity. Some modern cannons are equipped with stabilizer legs to stop the cannons from moving backwards (Fig 7.1n).

Fig 7.1m *A gun recoils when a bullet is fired.*

Fig 7.1n *A cannon with stabilizer legs.*

b Speed of air-rifle bullet

It is not easy to measure the speed of an air-rifle bullet directly as it is quite fast. Figure 7.1o shows a set-up that can be used to estimate its speed.

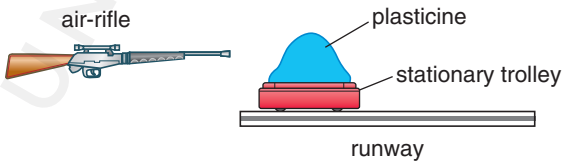


Fig 7.1o *Estimating the speed of air-rifle bullet.*

When a bullet is fired towards the plasticine, it will be embedded in the plasticine and move with the trolley after the collision. If the mass of the bullet, the mass of the trolley with plasticine, and the speed of the trolley after the collision are measured, the speed of the bullet can be calculated by applying the law of conservation of momentum.

The calculation is similar to that in Example 2 on p.287. ▶

Note the following precautions for a more accurate result.

1. The air-rifle should be close to the plasticine when firing the bullet.
2. The bullet should be fired along the moving direction of the trolley.
3. The runway should be friction-compensated. If a horizontal runway is used, the speed of the trolley **just after the collision** should be taken.

recoil 反衝

Simulation
Video

→ This simulation shows a Newton's cradle. Ss can pull the balls to one or both sides and can investigate both elastic and inelastic collisions.

→ This video shows a demonstration using a Newton's cradle.

C Newton's cradle

Newton's cradle is a toy consisting of a line of steel balls of equal mass (Fig 7.1p). When some balls are released on one side, the same number of balls rise to the same height on the other side after the collision. It can demonstrate the conservation of momentum as well as the conservation of energy in collision.

A small amount of energy is dissipated in each collision and the balls eventually come to rest.

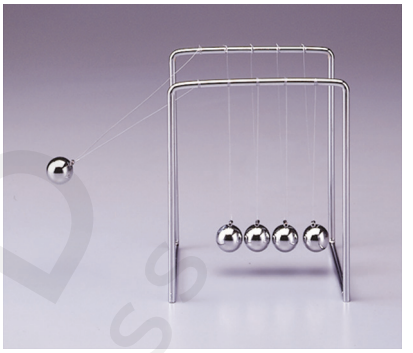


Fig 7.1p *Newton's cradle.*

Know more

Explanation for the Newton's cradle

Assume that a Newton's cradle consists of five identical balls each of mass m.

Question 1

Ball A is pulled to one side and released from height h (Fig a). After collision, A stops and E rises to height h on the other side. How does this demonstrate the conservation of momentum?

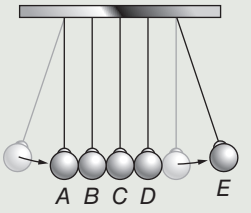


Fig a

Solution

By the law of conservation of energy, E rising to the same height as the initial position of A implies that they have the same speed at their lowest points. Since both A and E move in the same direction at the lowest point, their momenta at the lowest point are the same. Therefore, the total momentum of the balls is conserved in the collision.

Question 2

A and B are pulled to one side and released together (Fig b). They collide with other balls at speed u and stop. Is it possible for E to move alone on the other side after the collision?

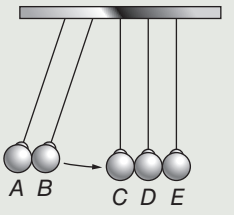


Fig b

Solution

Suppose only E moves just after collision and its velocity is v.

By the law of conservation of momentum,

$2mu = mv$

$v = 2u$

Total KE just before collision $= \frac{1}{2} \times 2m \times u^2 = mu^2$

Total KE just after collision $= \frac{1}{2} \times m \times (2u)^2 = 2mu^2 > mu^2$

This violates the law of conservation of energy.

$\therefore$ E will not move alone on the other side after the collision.

5 Momentum not conserved?

When you start walking, you gain momentum while the Earth seems to remain at rest (Fig 7.1q). The total momentum of the system (you and the Earth) does not seem to be conserved. Why? Let us try the following activity to find the answer.

Fig 7.1q *The Earth seems to remain at rest when you walk.*

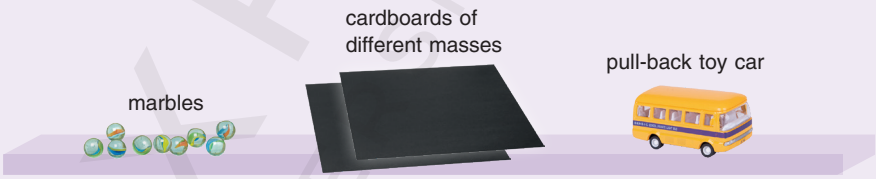
Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Is momentum conserved?

What you need



What to do

1. Put the cardboard on the marbles. Pull back the toy car on the table and allow it to move on the cardboard.
2. Repeat using another cardboard of greater mass.

What you find

How do the toy car and the cardboard move?

In the above activity, the toy car and the cardboard move in opposite directions. This complies with the law of conservation of momentum. Moreover, the more massive the cardboard is, the slower it moves.

The total momentum of you and the Earth is indeed conserved, but the mass of the Earth is so huge that its motion is not noticeable.

Try Q5 of Exercise 7.1 on p.299.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 An old cannon has a very massive base. Which of the following is the reason?

A This increases the momentum of the cannonball when firing.

B This reduces the momentum of the cannon when it fires a cannonball.

C This reduces the recoil speed of the cannon when it fires a cannonball.

D This increases the recoil speed of the cannon when it fires a cannonball.

2 You walk forwards on the ground but the Earth seems not to move backwards because

A the Earth and you are not in the same straight line.

B the Earth is too massive that its change in velocity is negligible.

C the law of conservation of momentum does not hold for the Earth.

3 Ball *X* of a Newton's cradle is released from one side (Fig a). What will happen after *X* collides with the other balls?

A Only one ball will move up on the other side.

B The other four balls will move up on the other side.

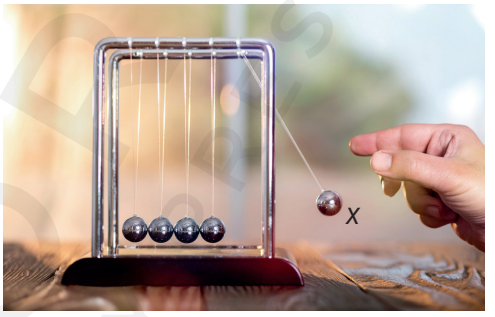


Fig a

Exercise 7.1

1 In which of the following types of collisions is the total KE of the colliding objects **not** conserved?

(1) Elastic collision

(2) Inelastic collision

(3) Completely inelastic collision

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

2 Two balls collide head-on. No external net force acts on the system of the two balls and the collision is elastic. Which of the following statements is/are correct?

(1) After the collision, the two balls may move at the same velocity.

(2) Sound may be generated during the collision.

(3) During the collision, there is a net force acting on each ball.

A (2) only B (3) only

C (1) and (3) only D (1), (2) and (3)

***3** A rocket of mass *m* moves at a constant velocity *v* in outer space. At time *t*, part *X* is ejected backwards and the remaining part *Y* moves forwards.

Fig a

The total momentum of *X* and *Y* just after time *t*

A is smaller than mv. B is equal to mv.

C is larger than mv. D cannot be determined.

***4** The magnitude of the momentum and the kinetic energy of runner *X* are *p* and *E* respectively. If runner *Y* has the same mass as *X* but runs twice as fast, what are the magnitude of the momentum and the kinetic energy of *Y*?

	Momentum	Kinetic energy
A	$2p$	$2E$
B	$2p$	$4E$
C	$4p$	$2E$
D	$4p$	$4E$

5 Assume the Moon is a stationary ball of mass 7.34×10^{22} kg. An astronaut of mass 90 kg starts walking from rest on the Moon. He accelerates to 1.5 m s^{-1} after 1 s. Estimate how fast the Moon moves due to his motion at this instant.

6 Goal! A football flies into the net (Fig b)!

Fig b

Is the total momentum of the ball and the net conserved in this collision? Explain your answer.

No

7 An astronaut is floating and remains at rest in a spaceship in outer space. She throws a box of mass 2 kg forwards by applying a constant force on it for 0.8 s. After the box has left her hands, it moves at 0.3 m s^{-1} forward and she moves at 0.01 m s^{-1} backwards. Assume that no external net force acts on her and the box. Find her mass.

***8** Boat A of mass 50 kg and boat B of mass 40 kg collide head-on. Their velocities before and after the collision are as shown (Fig c). Assume that no external net force acts on the boat system.

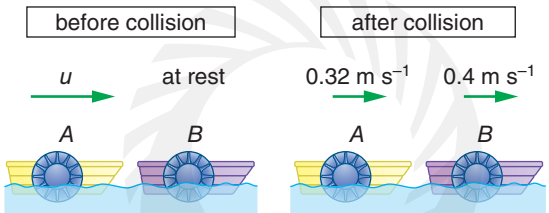


Fig c

(a) Compare the momenta of A and B after the collision.

(b) Compare the KE of A and B after the collision.

(c) Find u.

(d) Find the change in total KE of the boats.

***9** Two identical balls X and Y collide head-on. Their velocities before the collision are shown in Figure d.

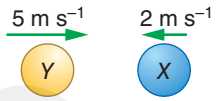
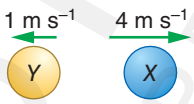
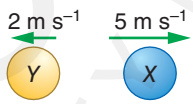
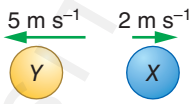


Fig d

Assume no external net force acts on the system of the two balls. If the collision is **elastic**, determine whether each figure below shows the possible velocities of X and Y after the collision. Explain your answer with calculation.

(a) 

(b) 

(c) 

***10** Eddie sits in a sledge which moves at 3 m s^{-1} towards the right. The total mass of Eddie and the sledge is 45 kg. Sally drops a snowball of mass 5 kg vertically onto the sledge (Fig e). It is known that the snowball is moving downwards at speed 2 m s^{-1} just before the collision.

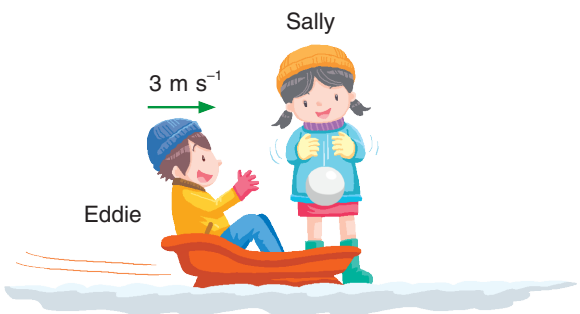


Fig e

(a) By considering the momentum along the horizontal direction, find the speed of the sledge just after the snowball has landed on it.

(b) Find the loss in the total KE of Eddie and the sledge during the process.

(c) Find the gain in KE of the snowball during the process.

7.2 Change in momentum

In the last section, we say that the total momentum of a system is conserved when there is no external net force acting on the system. Now we are going to discuss how the momentum of an object changes when it is acted on by a net force.

DSE goal

Interpret force as the rate of change of momentum (Newton's second law of motion).

1 Newton's second law of motion and change in momentum

Consider an object of mass m moving along a straight line. A constant net force F_{net} acts on the object along its direction of motion. The velocity of the object changes from u to v in time t (Fig 7.2a).

Fig 7.2a *A net force F_{net} acting on an object for time t.*

By Newton's second law,

$$F_{net} = ma = m \times \frac{v - u}{t} = \frac{mv - mu}{t}$$

Here mu is the object's initial momentum and mv is its final momentum. So $mv - mu$ is the change in momentum.

$$\textbf{net force} = \frac{\textbf{change in momentum}}{\textbf{time taken}}$$

$$F_{net} = \frac{mv - mu}{t} \quad \text{..... (1)}$$

Therefore, Newton's second law of motion can be restated as:

The net force acting on an object is equal to the rate of change in momentum of the object.

In fact, this is how Newton stated his second law of motion in his book *Principia*. He regarded momentum as a quantity of motion that is changed by the net force. We can do an experiment to verify this relationship.

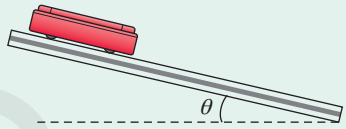
Keep a safe distance from the runway and trolleys.

Video

→ This video demonstrates Expt 7e.

Experiment 7e Net force and change in momentum

Measure the mass m of a trolley. Put the trolley on a runway inclined at angle θ to the horizontal. Start data-logging and release the trolley from rest. Measure its velocity v after it has travelled for a certain time t. Calculate the component $W_{//}$ of its weight along the runway by $W_{//} = mg \sin \theta$. It can be taken as the net force F_{net} acting on the trolley.



Precaution

Lubricate the wheels of the trolley to minimize friction.

Discussion

Is the net force acting on the trolley equal to the rate of change in momentum?

Rearrange equation (1) on p.300, we get

$$F_{net}t = mv - mu$$

This means that **a net force acting on an object for a period of time results in a change in its momentum**.

Example 5 Force on the ejected gas

A rocket car ejects 30 kg of hot gas at 500 m s^{-1} backwards in 1 s when it starts from rest. Find the force acting on the gas by the car during this period of time. Assume that this force is constant.

Solution

Take the direction of ejection of the gas as positive.

$$F_{net} = \frac{mv - mu}{t} = \frac{30 \times 500 - 0}{1} = 15\,000 \text{ N}$$

The force acting on the gas by the car is 15 000 N.

Think about...

What is the forward force acting on the rocket car by the gas?

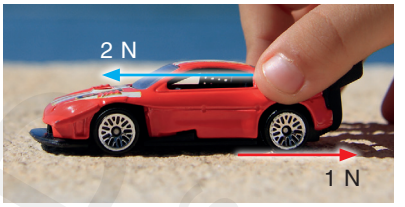
In fact, the equation $F_{net} = \frac{mv - mu}{t}$ can be applied even when the net force is not constant. In this case, F_{net} is the average net force acting on the object.

Tip

The force acting on the gas by the car is the net force on the gas.

Example 6 | Change in momentum

Tommy pushes his toy car, which is initially at rest, with a constant horizontal force of 2 N towards the left. The friction acting on the toy car is 1 N. Find the momentum of the toy car after 0.5 s.



Solution

Take the direction towards the left as positive.

By $F_{net}t = mv - mu$,

$mv = F_{net}t + mu = (2 - 1) \times 0.5 + 0 = 0.5$ kg m s^{-1}

The momentum of the toy car after 0.5 s is 0.5 kg m s^{-1} (towards the left).

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 The net force acting on an object is equal to

A its change in momentum. B its change in KE.

C its rate of change in momentum. D its rate of change in KE.

2 A net force F_{net} acts on an object of mass m for time t. The object moves along a straight line and its velocity changes from u to v. Complete the table below.

	F_{net} / N	t / s	m / kg	u / m s^{-1}	v / m s^{-1}
(a)		0.3	2	0	0.75
(b)	8		4	−10	−6

Learn by practice 3

The objects below move along the horizontal direction. Take the direction towards the right as positive.

1 An object's momentum changes from −2 kg m s^{-1} to 2 kg m s^{-1} in 5 s. Find the net force acting on it during this period of time.

2 A horizontal net force acts on an object of mass 2 kg for 3 s. During this period of time, the velocity of the object changes from 1 m s^{-1} to −2 m s^{-1}. Find the net force.

3 A net force of 10 N pointing towards the right acts on an object of mass 6 kg for 2 s. If the final velocity of the object is 7.5 m s^{-1}, find its initial velocity.

DSE goal

Understand that a net force acting on an object for a period of time results a change in momentum.

DSE exam

13(1B)Q6(b), 15(1A)Q6, 16(1B)Q4, 20(1A)Q5, 22(1A)Q7, 22(1A)Q8, 22(1B)Q3(b), 23(1B)Q3(d), etc

2 Impact

a Impact force

During an impact, the colliding objects exert **impact forces** on each other. These forces can be very large and may deform the objects (Fig 7.2b). We will study impact forces in the following experiment.

Fig 7.2b *High speed photo showing how a ball deforms during an impact.*

Video

→ This video shows the deformation of a baseball when it is hit by a bat.

Experiment 7f Investigating impact force

Keep a safe distance from the trolleys.

Video

→ This video demonstrates Expt 7f.

1 Set up the apparatus as shown with a rubber bumper attached to the trolley.

2 Start data-logging. Push the trolley so that it moves along the horizontal runway and its force sensor collides with a barrier. Note the velocity–time graph and force–time graph obtained.

Discussion

1 Is the impact force constant?

2 Compare the area under the force–time graph with the change in momentum of the trolley. What is the physical meaning of this area?

✎ Since the impact time is just about 0.015 s, a high data-sampling rate is recommended in Expt 7f to obtain a valid comparison between the area under the *F*–*t* graph and the change in *p*.

impact force 撞擊力

A motion sensor has a much lower sampling rate than a force sensor. Thus the *v*–*t* graph and *F*–*t* graph show different impact time.

Figures 7.2c and 7.2d show the results of the experiment. We can see in Figure 7.2d that the impact force acting on the trolley changes with time. The period of time in which the impact force is acting is called the **impact time**.

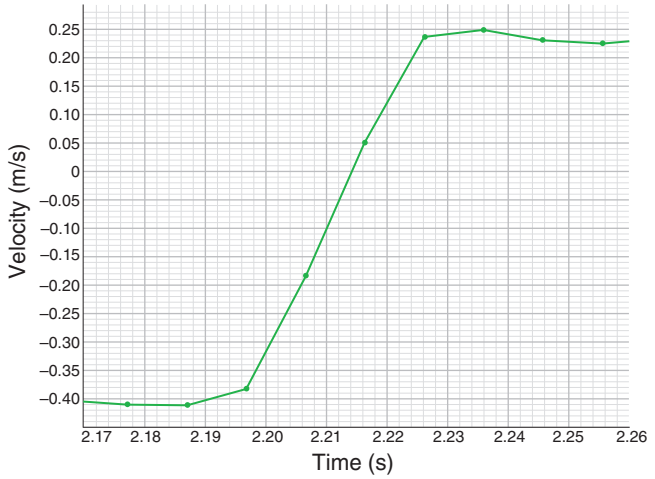


Fig 7.2c *A v–t graph obtained in Experiment 7f. Mass of trolley is 0.5 kg.*

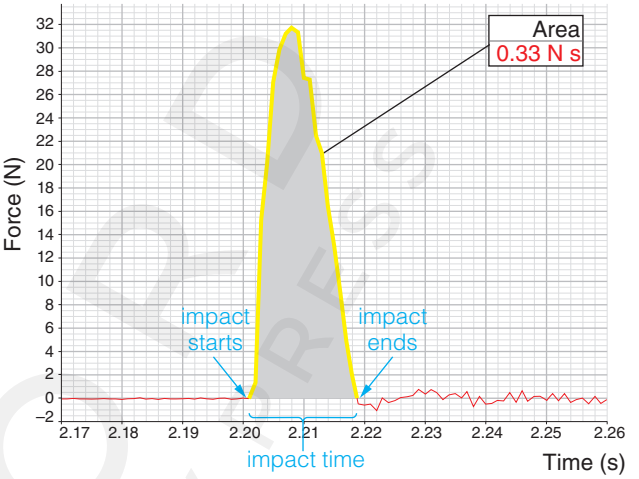


Fig 7.2d *An F–t graph obtained in Experiment 7f.*

The change of the size of an impact force can be observed in the case of hitting a baseball (Fig 7.2e). The ball deforms more as the impact force increases. It deforms most when the impact force reaches its maximum. Then the impact force decreases and the ball regains its shape.

Fig 7.2e *Deformation of a baseball changes as the impact force changes.*

b Force–time graph and change in momentum

Besides showing how the impact force F changes with time t, an F–t graph can also tell us how the momentum of an object changes on impact. Recall the equation $F_{net}t = mv - mu$. In an F–t graph **with F equal to the net force** acting on the object, $F_{net}t$ is the area under the graph. Therefore,

the area under *F*–*t* graph = change in momentum

This is similar to the relationship:
area under *v*–*t* graph = displacement *s*

impact time 撞擊時間

This relationship holds even if F is not constant (Fig 7.2f). The unit of the area under an F–t graph is the newton second (N s), which is equivalent to the unit of momentum (kg m s^{-1}).

The results shown in Figures 7.2c and 7.2d on p.314 give an example. The area under the graph in Figure 7.2d is 0.33 N s. This is equal to the change in momentum obtained in Figure 7.2c.

change in momentum
$= mv - mu$
$= 0.5 \times 0.25 - 0.5 \times (-0.41)$
$= 0.33$ kg m s^{-1}

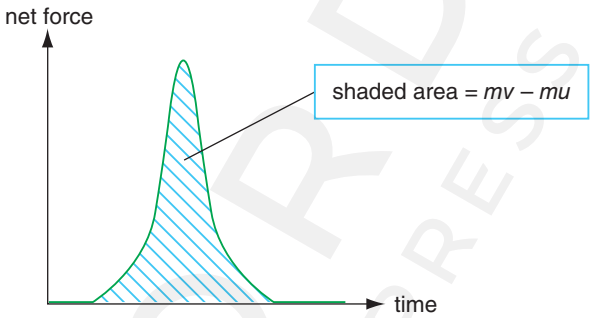
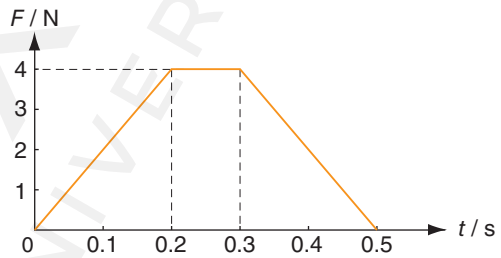


Fig 7.2f *Area under F–t graph is equal to change in momentum.*

Example 7 | *F–t* graph and average force

The figure shows how the net force F acting on a ball varies with time t.



In this period of time,

(a) what is the maximum net force acting on the ball?

(b) what is the change in momentum of the ball?

(c) what is the average net force acting on the ball?

Solution

(a) 4 N

(b) Change in momentum = area under F–t graph

$$= \frac{(0.1 + 0.5) \times 4}{2}$$

$$= 1.2 \text{ kg m s}^{-1}$$

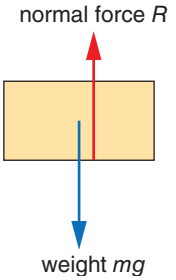
(c) Average net force $= \dfrac{\text{change in momentum}}{\text{time}} = \dfrac{1.2}{0.5} = 2.4$ N

Example 8 | Impact force on a falling egg

An egg of mass 0.05 kg falls vertically onto the ground. It moves at 4 m s^{-1} just before hitting the ground. It does not rebound after the impact. The impact time is 4 ms. Find the average impact force acting on the egg by the ground.

Solution

Take the upward direction as positive.



Average net force $F_{net} = \frac{mv - mu}{t} = \frac{0 - 0.05 \times (-4)}{4 \times 10^{-3}} = 50$ N

Since $F_{net} = R - mg$,

$R = F_{net} + mg = 50 + 0.05 \times 9.81 = 50.5$ N

The average impact force is 50.5 N (upwards).

Tip

When an object hits a surface or another object vertically, the net force equals to the resultant of the impact force and its weight.

Q39 Exam Report: This question tested candidates' knowledge and understanding of Newton's laws of motion via an unfamiliar situation. Candidates' performance was poor.

(a) Few candidates knew that the momentum change of water requires a net force from the jetpack and as a result of an action-and-reaction pair an equal and opposite supporting force is produced. Many had a misconception that this force comes from the interaction between the water ejected from the jetpack and the water surface.

(b) More than half of the candidates managed to draw the correct free-body diagram.

(c) (i) Some had difficulty in applying $F = \Delta p/\Delta t$ to find F as the directions of water flow needed to be considered.

(ii) A few candidates did not know what mechanical energy consists of.

(d) Candidates' performance was unsatisfactory. Quite a number of them suggested that a greater power is required for the person to stay at a higher position.

Think about...

Assume the egg can withstand an impact force of 60 N without breaking. Will the egg break in the above case? Why?

C Controlling impact time

An egg is easily broken. However, if we throw the egg towards a cloth as shown (Fig 7.2g), the egg can remain unbroken. Do you know why?

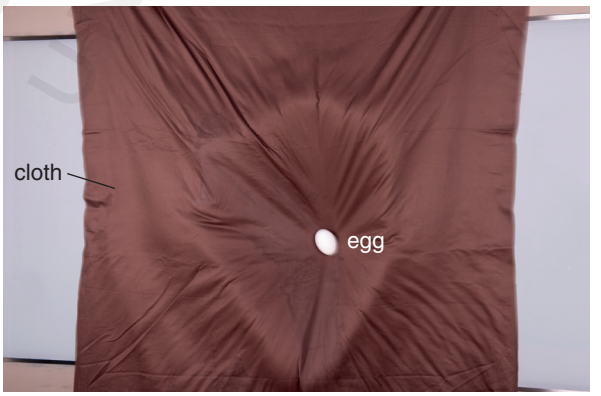


Fig 7.2g An egg remains unbroken after being thrown against a cloth.

When the egg collides with the cloth, the cloth deforms and therefore the impact time is increased. **According to *Ft = mv – mu*, since the change in momentum in this case is fixed, an increase in the impact time *t* will result in a decrease in the impact force *F*** acting on the egg. This greatly reduces the chance of breaking for the egg.

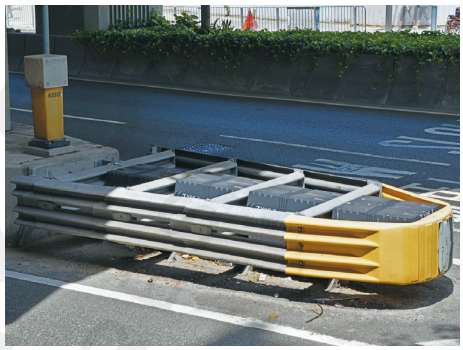
The change in momentum for stopping remains unchanged:

$\Delta p = mv - mu = 0 - mu = -mu$

✎ The following graphs best illustrate that a shorter impact time would result in a larger impact force for the same change in momentum.

A similar principle is applied in car safety. To reduce the impact forces acting on drivers and passengers in car accidents and hence help avoid serious injury, different methods are used to lengthen the impact time (Fig 7.2h).

(i) The front and rear ends of a car (the crumple zones) can deform in an accident.



(ii) A crash cushion system can deform if a car crashes into it.

Fig 7.2h *Methods to lengthen the impact time in car accidents.*

Know more

Reducing impact force

Devices that reduce impact forces by increasing the impact time can be found everywhere. Bubble wraps and foam are two common examples.

However, in some other cases, we may want to increase the time of impact to achieve a greater change in momentum. For example, a tennis player is encouraged to follow-through (Fig 7.2i) when striking a ball. This can increase the impact time between the racket and the ball, so the ball leaves the racket at a higher speed.

In this case, Δp of the ball is not fixed. Increasing the impact time can increase Δp and hence the final velocity v of the ball:

$$Ft = mv - mu$$

$$v = \frac{Ft + mu}{m}$$

Fig 7.2i *Follow-through motion in playing tennis.*

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

(For Q1–2.) When a tennis ball is hit by a racket, the net force acting on the ball is as shown (Fig a).

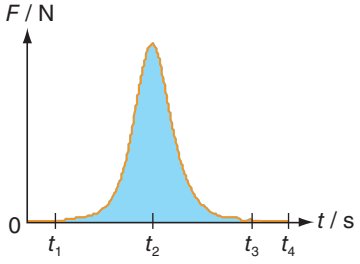


Fig a

1 What does the area under the curve represent?

A Work done by the racket during impact

B Rate of change in momentum of the ball

C Change in momentum of the ball

D Acceleration of the ball

2 At what time is the ball in contact with the racket?

A From 0 to t_4

B From t_1 to t_3 only

C At t_2 only

3 When an object of weight W falls vertically and hits the ground, the average impact force acting on the object by the ground is F. Which of the following gives the average net force on the object when hitting the ground?

A F

B W

C $F + W$

D $F - W$

Learn by practice 4

1 A tennis ball of mass 50 g flies towards a racket. After being hit by the racket, the ball bounces off at 6 m s^{-1} in the opposite direction. If the speed of the ball just before the impact is 4 m s^{-1} and the impact time is 0.2 s, find the average net force acting on the ball during the impact. Ignore the effect of gravity.

2 Michael bounces a basketball of 0.6 kg off the ground. The ball moves downwards at 0.7 m s^{-1} before the collision and moves upwards at 0.5 m s^{-1} after the collision. The impact time is 0.02 s.

(a) Find the average net force acting on the ball during the impact.

(b) Find the average impact force.

3 A 0.4 kg object is initially at rest at time $t = 0$. Figure a shows how the net force F_{net} acting on it changes with t.

(a) Find the impact time.

(b) Find the change in momentum of the object in the time period shown.

(c) Find the average net force on the object during the impact.

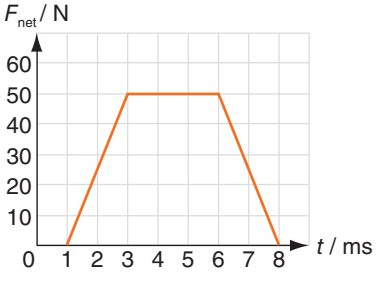


Fig a

Writing practice

Question

Drivers and passengers of cars are required by law to wear seat belts. A seat belt can stretch slightly during a sudden stop. Explain how this design protects the drivers and passengers. (2 marks)

Answer

As the seat belt stretches during a sudden stop, it lengthens the impact time, 1A

thus reduces the impact force on the drivers and passengers. 1A

Practise yourself

1 A glass dropped onto the hard floor will break, but one dropped onto a soft carpet will not. Explain the reason. (3 marks)

DSE goal

Relate the law of conservation of momentum to Newton's third law of motion.

3 Newton's third law of motion and law of conservation of momentum

When a rocket at rest in outer space ejects gas backwards, it moves forwards at the same time according to the law of conservation of momentum (Fig 7.2j).

We can also view it in this way. The rocket exerts a force on the gas to push it backwards. By Newton's third law of motion, the gas exerts a force on the rocket. This force propels the rocket forwards (Fig 7.2k).

This example shows that the law of conservation of momentum is closely related to Newton's third law.

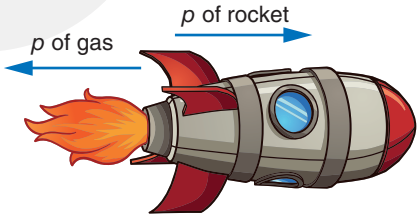


Fig 7.2j *Momenta of rocket and gas.*

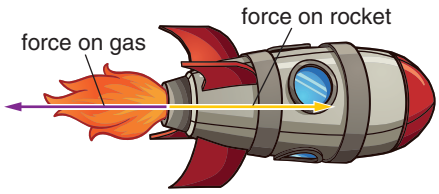


Fig 7.2k *Forces on rocket and gas.*

Consider two objects A and B that have a head-on collision on a smooth horizontal plane. The collision lasts for time t. Figure 7.2l shows the horizontal forces acting on the objects during the collision.

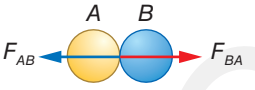


Fig 7.2l *Horizontal forces acting on A and B during collision.*

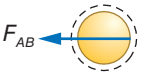


Fig 7.2m *Force acting on A.*

Fig 7.2n *Force acting on B.*

Consider A (Fig 7.2m). Force F_{AB} acts on A for time t, changing its velocity from u_A to v_A. As F_{AB} is the net force on A,

$$F_{AB} = \frac{m_A v_A - m_A u_A}{t}$$

Consider B (Fig 7.2n). Similarly, $F_{BA} = \dfrac{m_B v_B - m_B u_B}{t}$

Since F_{AB} and F_{BA} are an action-and-reaction pair,

$$F_{AB} = -F_{BA}$$

The negative sign means that F_{AB} and F_{BA} are in opposite directions.

$$\frac{m_A v_A - m_A u_A}{t} = -\frac{m_B v_B - m_B u_B}{t}$$

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

The equation can be rearranged as
$m_A v_A - m_A u_A = -(m_B v_B - m_B u_B)$
$\Delta p_A = -\Delta p_B$
This shows that in a system of two bodies, the changes in momentum (Δp) of the two bodies are equal in magnitude but opposite in direction.

This is the law of conservation of momentum.

4 Solving problems involving motion and collision along a straight line

Up to now, we have learned different methods to solve problems involving motion and collision along a straight line. In the following examples, a combination of these methods will be used.

Example 9 Falling object

A dry cell of mass 23 g falls from a height of 30 m (about the height of 10/F). When it collides with the ground, the impact time is 5 ms. Assume the collision is completely inelastic, i.e. the dry cell does not rebound, and the air resistance is negligible. Find the average force acting on the ground by the dry cell during the impact.

Solution

Take the downward direction as positive.

Consider the motion during the fall.

$u = 0$, $a = +9.81\ \text{m s}^{-2}$, $s = +30\ \text{m}$, $v = ?$

The law of conservation of energy can also be used to find v. ▶

By $v^2 = u^2 + 2as$,

$v^2 = 0^2 + 2 \times 9.81 \times 30$

$v = 24.26\ \text{m s}^{-1}$

During the impact,

$$F_{\text{net}} = \frac{mv - mu}{t} = \frac{0 - 0.023 \times 24.26}{0.005} = -111.6\ \text{N}$$

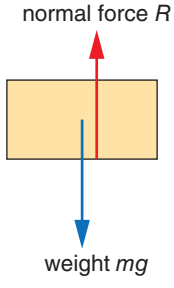
Since $F_{\text{net}} = mg - R$,

$-111.6 = 0.023 \times 9.81 - R$

$R = 112\ \text{N}$

By Newton's third law of motion, the average force acting on the ground by the dry cell during the impact is 112 N (downwards).

This is about 500 times the weight of the dry cell! It can be fatal if the cell hits a person. ▶



STSE

Danger of falling objects

As shown in Example 9, the impact force due to a falling object is very large. You should not allow any object to be dropped from your flat onto the street.

Example 10 Motion of a bouncing ball

A ball of mass 0.2 kg is released from rest below a motion sensor (Fig a). It hits the ground and bounces several times. Figure b shows the ball's v–t graph, with downwards taken as positive.

Ignore the size of the ball. Assume the ball only moves along the vertical direction.

Fig a

Ts may tell Ss that v is positive as the ball moves downwards (e.g. *AB* and *EF*) and negative as upwards (e.g. *DE*).
The motion between points *B* and *D* is not a free fall motion since the ball is in contact with the ground.

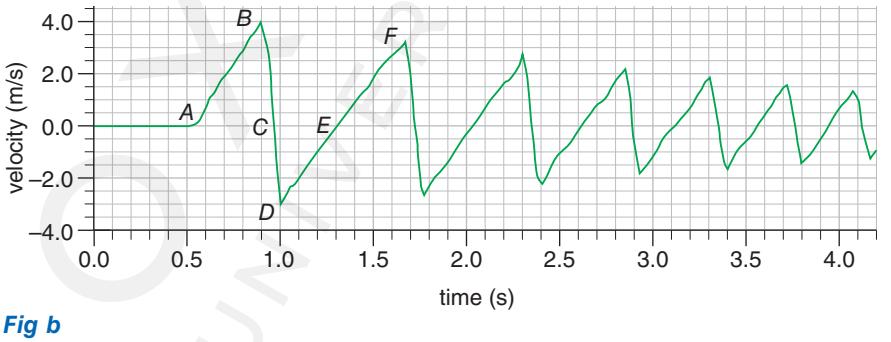


Fig b

(a) At what time does the ball
  (i) hit the ground the first time,
  (ii) reaches its maximum height during its first rebound?

(b) Describe the motion of the ball corresponding to the part from *D* to *F* in the graph.

(c) Estimate the acceleration due to gravity from the graph. Assume the air resistance is negligible.

(d) Estimate the average net force acting on the ball during its first impact with the ground.

(e) Is the total mechanical energy of the ball conserved during each impact? Explain your answer.

(f) Sketch the displacement–time graph of the ball from $t = 0$ to $t = 2.0$ s. No calculation is needed.

Tip
The velocity of the ball drops abruptly when it hits the ground.

Tip
The velocity of the ball is zero when it reaches its maximum height.

Solution

(a) (i) At $t = 0.9$ s

(ii) At $t = 1.3$ s

(b) From D to E, the ball moves upwards and slows down.

At E, it reaches its maximum height and is momentarily at rest.

From E to F, it speeds up from rest and moves downwards.

(c) During the fall from $t = 0.5$ s to $t = 0.9$ s,

a = slope of v–t graph

$$= \frac{4.0 - 0}{0.9 - 0.5}$$

$$= 10 \text{ m s}^{-2}$$

The acceleration due to gravity is 10 m s^{-2} (downwards).

(d) During the impact from $t = 0.9$ s to $t = 1.0$ s,

$$F_{net} = \frac{mv - mu}{t}$$

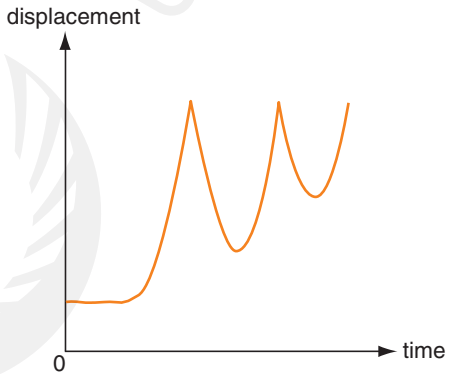
$$= \frac{0.2 \times (-3.0) - 0.2 \times 4.0}{0.1}$$

$$= -14 \text{ N}$$

The average net force acting on the ball during its first impact is 14 N (upwards).

(e) No. The ball has a lower speed after an impact than before the impact, showing that some KE has been lost.

(f)



☆ HKDSE 2016 Paper 1B Q4

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Momentum of water along a hose

Water flows through a hose on horizontal ground (Fig a). A gardener bends the hose so that water flows towards the right when entering the hose and flows towards the left when leaving it (Fig b).

Fig a

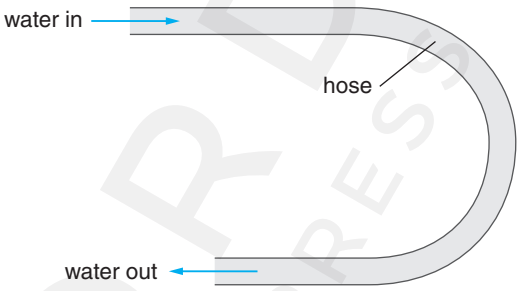


Fig b

Suppose the water enters and leaves the hose at the same speed of 2 m s^{-1} and 0.5 kg of water flows out of the hose in each second.

(a) Is the momentum of the water conserved when flowing through the hose? Explain. (2 marks)

(b) The water exerts a force towards the right on the hose when flowing through it at the turning point. Explain this using Newton's laws of motion. (3 marks)

(c) Find the force acting on the hose by the water. (2 marks)

Solution

(a) No, 1A

because the direction of the velocity of water has changed. 1A

(b) By Newton's second law of motion, the hose exerts a force on the water to change its momentum. 1A

As the hose exerts a force towards the left on the water, 1A

by Newton's third law of motion 1A

the water exerts a force towards the right on the hose.

Common mistake
Students may not know that the momentum change of water requires a net force acting on the water.

(c) Take the direction towards the left as positive.

Consider the force F acting on the water by the hose.

$$F = \frac{mv - mu}{t} = \frac{m}{t} \times (v - u) \quad \text{1M}$$

$$= 0.5 \times [2 - (-2)]$$

$$= 2 \text{ N}$$

Common mistake
Students may miss out the direction of water flow in their calculation.

The force acting on the water by the hose is 2 N (leftwards).

By Newton's third law of motion, the force acting on the hose by the water is 2 N (rightwards). 1A

Revision 7 Q39 (p.328)

Physics of car safety

Crumple zone

The front and rear of a car can collapse in a crash. This increases the impact time and reduces the impact force acting on the people in the car.

Air bag

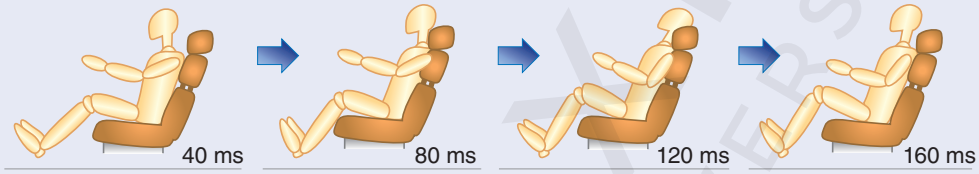
During a crash, the air bags are inflated within fractions of a second. They protect the people from hitting hard objects in the car. They also prolong the time for the people to come to a stop.

Seat belt

When a car stops suddenly, the people inside continue to move forwards due to inertia. Seat belts stop them from moving forwards so that they will not be thrown out of the car. Also, seat belts can stretch slightly to increase the impact time.

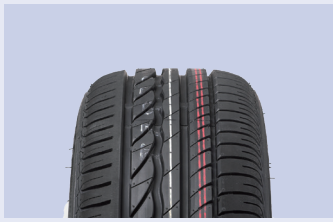
Seat belts reduce serious injuries and deaths in car accidents by about a half.

Head-rest



If a car is bumped into at the back by another car, the people inside is pushed forwards but their heads tend to remain at rest. Head-rests can protect them from neck injuries.

Tyre

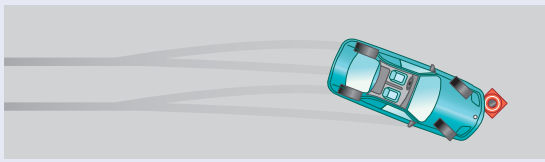


The friction between the tyres and the ground makes a car move. The tread pattern on tyres is for squeezing water out to ensure that the tyres have a firm grip on the ground.

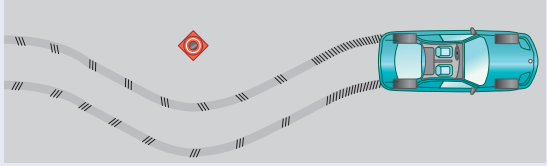
Anti-lock braking system (ABS)

A driver may step on the brake pedal hard in emergency braking. The wheels become locked up and the car skids on the road. The driver loses control and this can be very dangerous.

ABS automatically releases the brake slightly when the speed of any wheel is very different from the others. As a result, the brakes are applied and released repeatedly during braking so that the wheels are never locked up. The driver is in control and can steer the car away from danger.



The car skids and is out of control.



The ABS prevents the wheels from being locked up, so the car can be steered away from danger.

Exercise 7.2

1 A ball of mass 0.5 kg falls vertically. It hits the ground at 8 m s^{-1} and rebounds vertically upwards at 6 m s^{-1}. What is the change in momentum of the ball in the collision?

A 1 kg m s^{-1} (upwards)

B 1 kg m s^{-1} (downwards)

C 7 kg m s^{-1} (upwards)

D 7 kg m s^{-1} (downwards)

2 A rocket engine is fixed horizontally to the ground in a test to find out the forward force it produces (Fig a).

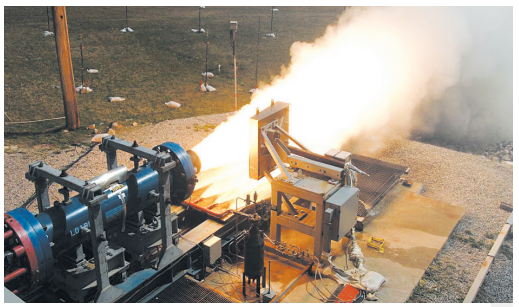


Fig a

If the engine ejects hot gas at 600 m s^{-1} and the forward force produced is found to be 1500 kN, how much gas does it eject each second?

A 400 kg

B 2500 kg

C 5000 kg

D 900 000 kg

*3 A wine glass can withstand an impact force of 50 N without breaking.

(a) A marble hits the glass and the average impact force is 45 N. Why does the glass still break?

(b) Explain why the glass can be protected from breaking by packing it with bubble wrap (Fig b).

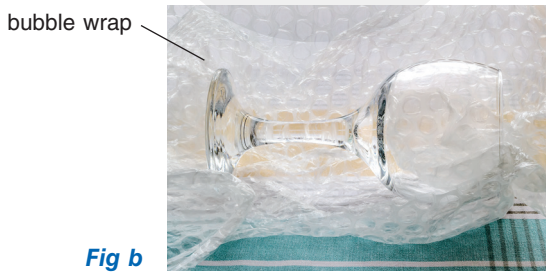


Fig b

*4 Object *X* collides head-on with object *Y* on a smooth horizontal plane. Show that the change in momentum of *X* has the same magnitude but is in the opposite direction to that of *Y* in the collision.

*5 A baseball of mass 145 g moves at 30 m s^{-1} towards the left just before it is hit by a bat. It flies towards the right just after being hit. Figure c shows how the force *F* acting on the baseball by the bat varies with time *t*.

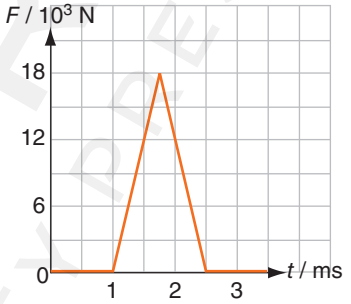


Fig c

(a) Find the change in momentum of the ball in the collision.

(b) Find the average force acting on the ball by the bat.

(c) Find the speed of the ball just after being hit.

*6 A cart moves along a straight line. Its momentum *p* changes with time *t* as shown (Fig d).

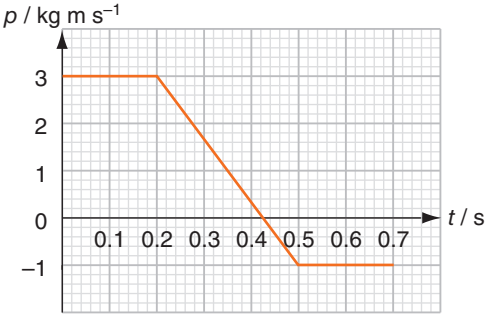


Fig d

Recall that the net force acting on an object is equal to the rate of change in momentum of the object. Consider the time period t = 0.2 s to t = 0.5 s.

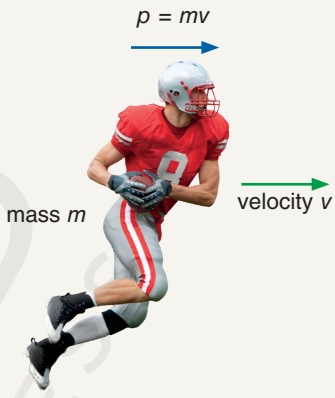
(a) Find the average net force on the cart.

(b) Is the net force on the cart during this time period constant? Explain.

Key concepts 7

1 Momentum p

- a quantity of motion
- $p = mv$
- unit: kg m s^{-1}
- vector (same direction as v)

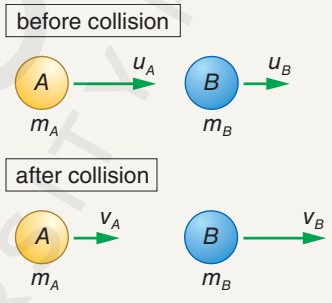


2 Law of conservation of momentum

no external net force ⇒ total p of a system is conserved

- external force
  - force exerted by objects outside the system
  - affects the total p of a system
- internal force
  - force exerted by objects within the system
  - does not affect the total p of a system

- for a system of 2 objects, with no external net force, $m_A u_A + m_B u_B = m_A v_A + m_B v_B$



	Total p conserved?	Total KE conserved?
Completely inelastic collision	✓	✗
Inelastic collision	✓	✗
Elastic collision	✓	✓
Explosion	✓	✗

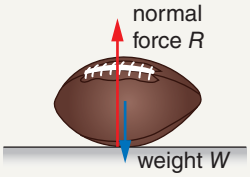
3 Change in momentum

- net force = rate of change in p

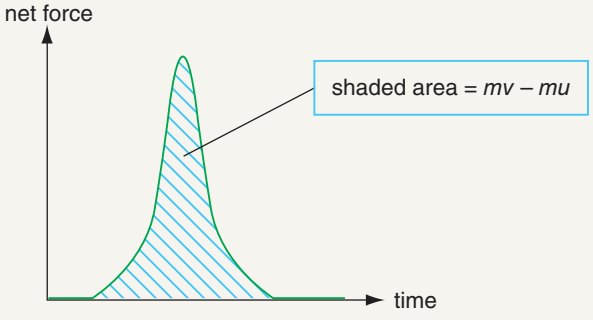
$$F_{net} = \frac{mv - mu}{t}$$

$$\Rightarrow F_{net}t = mv - mu = \Delta p$$

- for fixed Δp, $t \uparrow \Rightarrow F_{net} \downarrow$
- if Δp is not fixed, $t \uparrow$ or $F_{net} \uparrow \Rightarrow \Delta p \uparrow$
- when an object hits the ground, net force $F_{net} = R - W$ = rate of change in p



- during an impact, the net force changes with time.



- area under force–time graph = Δp

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Key terms

1 collision 碰撞 p.279
2 elastic collision 彈性碰撞 p.289
3 impact force 撞擊力 p.303
4 impact time 撞擊時間 p.304
5 inelastic collision 非彈性碰撞 p.289
6 law of conservation of momentum 動量守恆定律 p.286
7 momentum 動量 p.279

Revision 7

Multiple-choice questions

★1 7.2 A pole vault athlete passes the bar with the aid of a pole (Fig a) and is about to land on a mat.

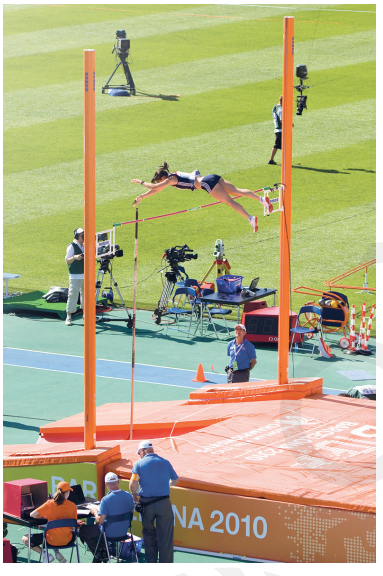


Fig a

Which of the following is/are the reason(s) why the mat can protect the athlete from injury?

(1) The mat reduces the change in momentum of the athlete during impact.
(2) The mat reduces the rate of change in momentum of the athlete during impact.
(3) The mat deforms when the athlete lands on it.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

★2 7.2 Baseballs are thrown towards a player and the player hits the balls back (Fig b). In which of the following cases is the average impact force acting on the ball the largest? Assume all balls move in the opposite direction after the impact.

Fig b

	Initial speed of the ball	Rebound speed of the ball	Impact time
A	110 km h^{-1}	96 km h^{-1}	0.0013 s
B	120 km h^{-1}	98 km h^{-1}	0.0018 s
C	130 km h^{-1}	112 km h^{-1}	0.0021 s
D	140 km h^{-1}	110 km h^{-1}	0.0013 s

★3 7.2 A ball of mass 80 g falls from rest from a height 0.5 m above the ground. The ball bounces to a height 0.4 m above the ground after colliding with the ground. The impact between the ball and the ground lasts for 6 ms. What is the magnitude of the average net force acting on the ball during the impact?

A 4.41 N
B 5.19 N
C 79.1 N
D 79.9 N

★4 7.2 An object moves along a straight line. The following graph shows how its momentum changes with time (Fig c).

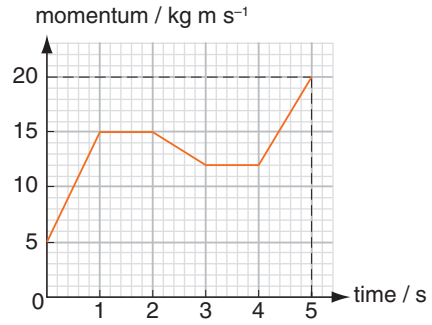


Fig c

What is the largest net force acting on the object?

A 3 N
B 5 N
C 8 N
D 10 N

★5 7.2 Snow cannons are used to produce snow on ski slopes (Fig d). A snow cannon ejects 60 kg of snow per minute. Suppose the snow ejected by the cannon strikes a tree horizontally at 10 m s^{-1} and stops. Find the magnitude of the average force exerted by the snow on the tree.

Fig d

A 6 N
B 10 N
C 36 N
D 600 N

★6 7.1 A cyclist of mass 60 kg decelerates at 2 m s^{-2} for 3 s along a straight line. His final momentum is 150 kg m s^{-1}. What is his initial momentum?

A 110 kg m s^{-1}
B 190 kg m s^{-1}
C 360 kg m s^{-1}
D 510 kg m s^{-1}

★7 7.1 Two bumper cars *X* and *Y* are travelling towards the right. *X* is travelling at 1.5 m s^{-1} while *Y* is travelling at 1.1 m s^{-1} when they collide. The velocities of *X* and *Y* are 1.1 m s^{-1} and 1.6 m s^{-1} towards the right after the collision (Fig e). What is the ratio of the mass of *X* to that of *Y*? Assume that the external net force acting on the bumper car system is negligible.

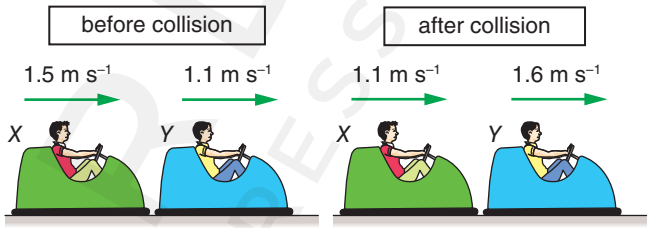


Fig e

A 1 : 0.8
B 1 : 1
C 1 : 1.17
D 1 : 1.4

★8 Misc Monkey M_1 holds a rattan 5 m long and swings from position *X*. It collides with a tree when the rattan is vertical (Fig f). Another monkey M_2 of the same mass falls from a height of 5 m and collides with the ground (Fig g).

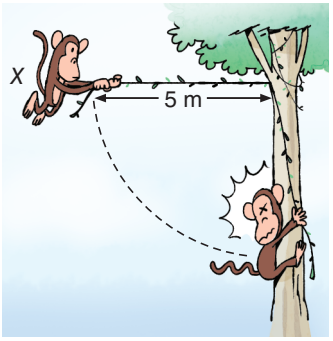


Fig f

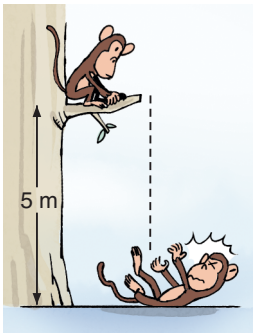


Fig g

Assume the impact time in both cases is the same and both collisions are completely inelastic. Which monkey experiences a larger impact force?

A M_1
B M_2
C They experience impact force of the same magnitude.
D Cannot be determined as the mass of the monkeys is not known.

★9 7.2 An egg falls on some cotton wool and stops. The graph below shows how the net force F acting on it changes with time t (Fig h).

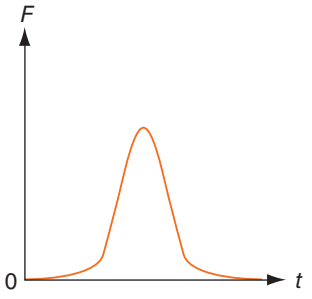
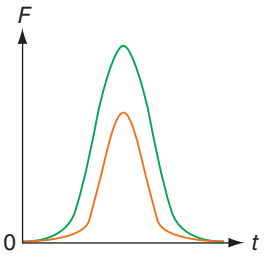


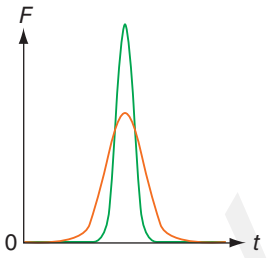
Fig h

Which of the following graphs in green best shows how F varies with t if the egg falls on the hard floor instead of the cotton wool?

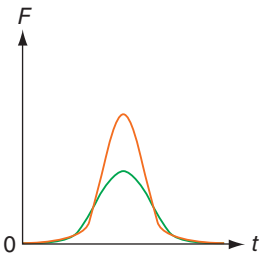
A



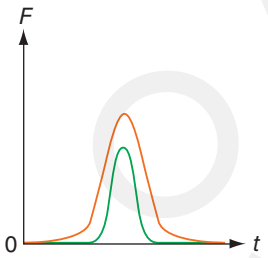
B



C



D



★10 7.1 Object X collides head-on with object Y which is initially at rest, on a smooth horizontal plane. The mass of X is m and it moves at a speed u before the collision. The mass of Y is M and it moves at a speed v after the collision. Which of the following statements is/are correct?

(1) Mv can be larger than mu.

(2) v can be larger than u.

(3) Mv^2 can be larger than mu^2.

A (2) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

★11 Misc Cart X is initially at rest on the horizontal part of a smooth track (Fig i). Cart Y moving at velocity v collides with X and they stick together after collision. The two carts have the same mass. Ignore their sizes.

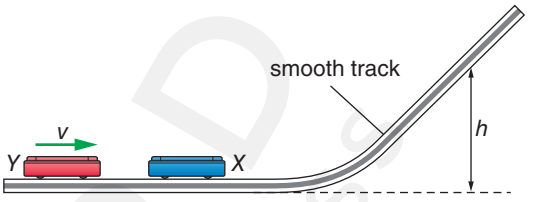


Fig i

What is the maximum height h that the carts can reach?

A $\frac{v^2}{8g}$ B $\frac{v^2}{4g}$

C $\frac{v^2}{2g}$ D $\frac{2v^2}{g}$

★12 Misc Figure j shows a garden sprinkler ejecting water. Figure k shows the top view of the sprinkler, which consists of two tubes X and Y. Water is ejected from each tube at 20 g s^{-1}. The water enters each tube at 2 m s^{-1} and leaves it at the same speed.

Fig j

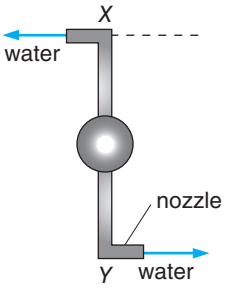
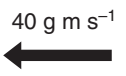


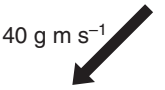
Fig k

Which diagram represents the change in momentum of water in tube X?

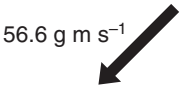
A



B



C



D

*13 Misc Objects X and Y move in the same direction and have the same KE. The mass of X is twice that of Y. If the momentum of X is p, what is the momentum of Y?

A $\frac{p}{2}$

B $\frac{p}{\sqrt{2}}$

C $\sqrt{2}p$

D $2p$

*14 Misc A stationary object explodes into a lighter part and a heavier part. Which of the following statements must be correct?

A The lighter part possesses more KE than the heavier part.

B The lighter part possesses the same amount of KE as the heavier part.

C The lighter part has a lower speed than the heavier part.

D The lighter part has the same speed as the heavier part.

*15 Misc Astronaut Y of mass 160 kg floats away at 0.2 m s^{-1} towards the right after the safety cord connecting him to a space station breaks. Astronaut X of mass 180 kg immediately chases Y at 0.4 m s^{-1} (Fig l). X catches Y and they move together after meeting. Then X uses the device on his back to eject gas at 50 m s^{-1} at a rate of 20 g s^{-1}. To which direction and for how long should he eject the gas to stop moving?

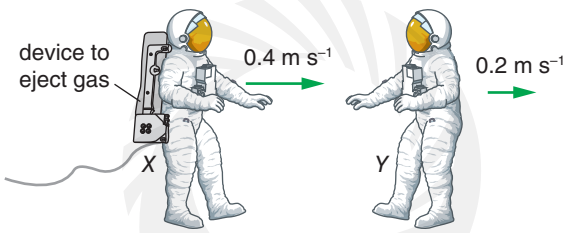


Fig l

A 40 s (towards left)

B 40 s (towards right)

C 104 s (towards left)

D 104 s (towards right)

*16 7.1 A water tank is put on a trolley (Fig m). It is given a sharp push and moves along a straight line on a smooth horizontal plane. Raindrops are falling vertically into it at a rate of 100 g min^{-1}. At time t = 0, the total mass of the tank and the trolley is 500 g and their speed is 2 m s^{-1}. What is their speed at t = 90 s?

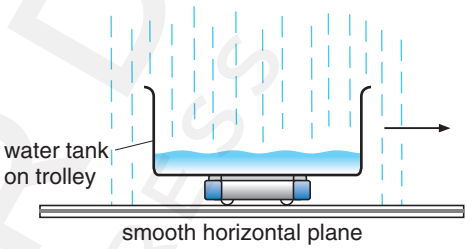


Fig m

A 1.54 m s^{-1} B 1.67 m s^{-1}

C 2 m s^{-1} D 6.67 m s^{-1}

**17 7.2 A boy throws a ball vertically upwards in a room. The ball hits the ceiling at 2 m s^{-1} and rebounds at 2 m s^{-1} downwards (Fig n). The mass of the ball is 0.1 kg. The impact time is 0.08 s. What is the magnitude of the average force acting on the ball by the ceiling during the impact?

A 1.88 N

B 4.02 N

C 5 N

D 5.98 N

Fig n

Refer p.306

**18 Misc Carts X and Y have the same mass of 0.5 kg. They move in the same direction along a smooth horizontal straight track. Initially, X moves at 2 m s^{-1} and Y moves at 1 m s^{-1} (Fig o). X moves at 1 m s^{-1} towards the right after colliding with Y. What is the work done on Y by X?

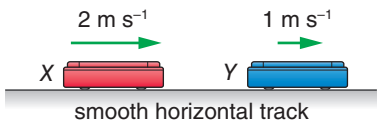


Fig o

A 0.25 J B 0.5 J

C 0.75 J D 1 J

Refer p.255

**19 Misc

Two astronauts X and Y play ball in a spacecraft in outer space (Fig p). The mass of each astronaut is 60 kg and the mass of the ball is 0.125 kg. The astronauts and the ball are at rest initially. X passes the ball at 3 m s^{-1} to Y and Y returns it at 3 m s^{-1}. They keep passing the ball at the same speed. What is Y's speed after passing the ball for the 10th time? Assume that they keep moving along a straight line and no external net force acts on the system of the astronauts and the ball.

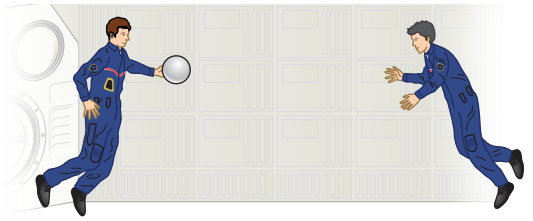


Fig p

A 0

B 0.0625 m s^{-1}

C 0.125 m s^{-1}

D 0.25 m s^{-1}

▶ Refer p.290

20 **HKDSE 2015 Paper 1A Q6**
7.2

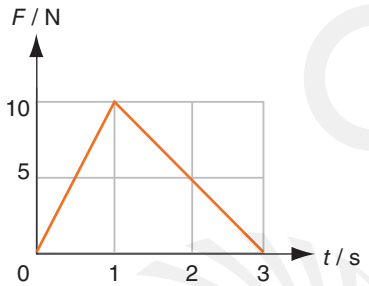


Fig q

An object of mass 3 kg is initially at rest on a smooth horizontal ground. A force F is applied horizontally to the object such that the magnitude of F varies with time t as shown. What is the speed of the object at t = 3 s? Neglect air resistance.

A 2.5 m s^{-1} (10%)

B 5 m s^{-1} (50%)

C 10 m s^{-1} (22%)

D 15 m s^{-1} (18%)

Q20 Exam Report: 50% of the candidates managed to use the graph to find the final speed of the object.

21 **HKDSE 2016 Paper 1A Q12**
Misc

On a smooth horizontal surface, a marble P moving with speed u collides head-on with another marble Q, which is at rest. After collision, P and Q move with different speeds as shown.

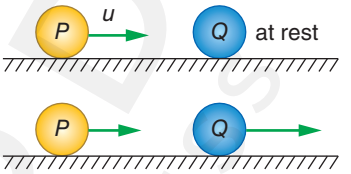


Fig r

Which of the following statements about this collision is/are correct?

(1) During collision, the force acting on Q by P is equal and opposite to that acting on P by Q.

(2) The total momentum of the two marbles is conserved only when the collision is perfectly elastic.

(3) The kinetic energy lost by P must be equal to that gained by Q.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

22 **HKDSE 2019 Paper 1A Q11**
7.1

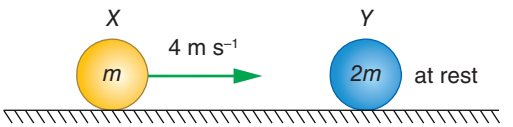


Fig s

On a smooth horizontal surface, sphere X of mass m travels with speed 4 m s^{-1}. It collides head-on with another sphere Y of mass $2m$, which is at rest initially. Which of the following can be the speed of Y just after collision?

(1) 1 m s^{-1}

(2) 2 m s^{-1}

(3) 3 m s^{-1}

A (1) only (12%)

B (2) only (27%)

C (1) and (2) only (41%)

D (2) and (3) only (20%)

Q22 Exam Report: Just over one-quarter of the candidates were able to obtain the correct answer.

Q23 Exam Report: Over one-third of the candidates did not think that statement (2) could be correct.

23 **HKDSE 2020 Paper 1A Q5**
7.2

Fig t

For a car travelling on a highway, which of the following statements about the safety design of the headrest is/are correct?

(1) As the headrest is soft, it can reduce the force exerted on the passenger's head during impact.

(2) It can minimise injury of the passenger when the car is struck by another one from behind.

(3) It can minimise injury of the passenger when the car brakes suddenly.

A (1) only (25%) B (3) only

C (1) and (2) only (52%) D (2) and (3) only (13%)

Conventional questions

★**24** 7.1 A bullet of mass 5 g flies at 300 m s^{-1} horizontally towards a pumpkin which is at rest on a smooth horizontal plane. The bullet penetrates the pumpkin and leaves it at 200 m s^{-1} horizontally (Fig u). The mass of the pumpkin is 4 kg. Neglect the effect of the content flying out of the pumpkin.

Fig u

(a) Is the momentum of the pumpkin conserved? Explain your answer. (2 marks)

(b) Estimate the speed of the pumpkin just after the bullet has left it. (2 marks)

★**25** 7.2 A gangster uses a machine gun to attack Superman. The gun fires 90 bullets, each of mass 5 g. The bullets move at 400 m s^{-1} and stop after hitting Superman. The impact time for each bullet is 10 ms.

(a) Find the magnitude of the average force acting on Superman by each bullet. (2 marks)

(b) If the bullets rebound instead of stopping after hitting Superman and the impact time remains unchanged, how would the answer to (a) change? Explain briefly. (2 marks)

★**26** 7.2 A ball of mass 0.2 kg moves to the left on a smooth horizontal surface and hits an object. The ball moves at 1.5 m s^{-1} towards the right after the impact. The net force F_{net} acting on it points to the right during the impact and varies with time t as shown (Fig v). The area under the graph is 0.7 N s.

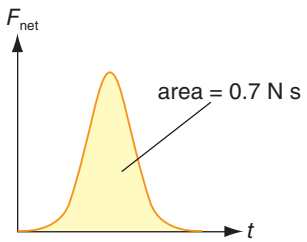


Fig v

(a) State the physical meaning of the area under the graph. (1 mark)

(b) Estimate the velocity of the ball before the collision. (2 marks)

(c) If the average net force acting on the ball during the impact is 17.5 N, estimate the impact time. (2 marks)

★**27** Misc A drone of mass 0.9 kg is shown in Figure w. When the drone remains stationary in the air, its four rotors push air downwards.

Fig w

(a) Explain briefly how the drone can remain stationary in the air. (3 marks)

(b) When the drone accelerates downwards at 0.2 m s^{-2}, 1.1 kg of air is pushed downwards at speed v each second. Assume that the air is initially at rest.

(i) Find v. (2 marks)

(ii) About 30% of the energy output by the drone becomes the KE of the air that is pushed downwards. Estimate the power output of the drone. (3 marks)

(iii) Will the power output be larger or smaller than that in (ii) when the drone remains stationary in the air? (1 mark)

★28 Misc A student carries out an experiment to verify the law of conservation of momentum. He pushes trolley A sharply towards stationary trolley B (Fig x). The mass of A is 0.5 kg and the mass of B is 1.5 kg.

Fig x

Figure y shows the velocity–time graphs of A and B. The direction towards the right is taken as positive.

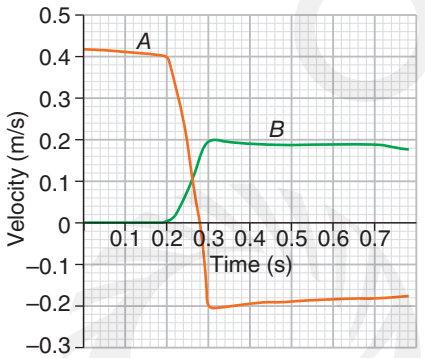


Fig y

(a) Show that the total momentum of the trolleys is conserved in the collision. (2 marks)

(b) Estimate the average force acting on A by B during the collision. (3 marks)

(c) Suggest one precaution for getting an accurate result. (1 mark)

★29 Misc Trolley X, with a spring attached to its front end, collides head-on with trolley Y on a smooth horizontal runway (Fig z). Before the collision, X moves at 0.9 m s^{-1} and Y moves at 0.6 m s^{-1} in the opposite direction. The mass of X is 0.5 kg and the mass of Y is 1 kg. The collision is elastic.

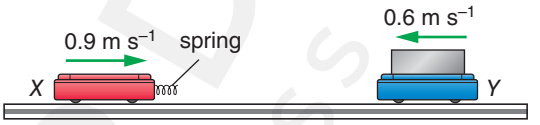


Fig z

(a) The spring is compressed during the collision. At the instant when the spring is compressed the most, the trolleys move at the same velocity.

(i) Find the common velocity of the trolleys. (2 marks)

(ii) Find the maximum elastic potential energy stored in the spring. (2 marks)

(b) David argues that as the collision is elastic, X will move to the left at 0.9 m s^{-1} and Y will move to the right at 0.6 m s^{-1} after the collision. Explain whether he is correct with his calculation. (3 marks)

★30 7.2 A driver not wearing a seat belt would be thrown forwards and would hit the windscreen if his car crashes. The net force that he would experience if he hits the windscreen is approximated in Figure aa.

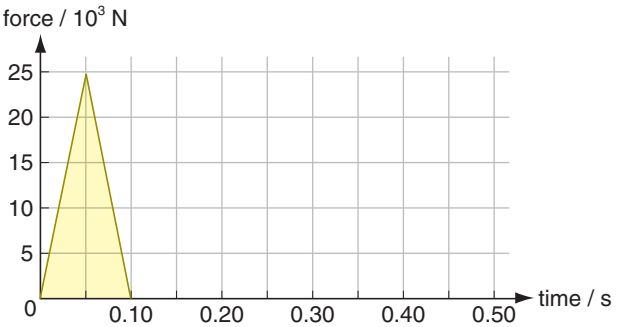


Fig aa

(a) Find the area under the graph. (1 mark)

(b) What is the physical meaning of the area in (a)? (1 mark)

(c) Find the average net force acting on the driver when he hits the windscreen. (2 marks)

(d) Suppose the driver wears a seat belt and the average net force that he would experience is reduced to $\frac{1}{5}$ during the crash.

(i) Explain how the seat belt reduces the force that the driver experiences. (2 marks)

(ii) In Figure aa, sketch the new F–t graph by approximating it as an isosceles triangle. (2 marks)

★**31** Misc An airsoft gun fires a bullet of mass 0.2 g towards a box which is hung by a string as shown (Fig ab). The bullet is embedded in the box after the collision. The box swings towards the right and the maximum height it reaches is 0.07 m above its original position. The mass of the box is 20 g. Assume the bullet hits the box horizontally.

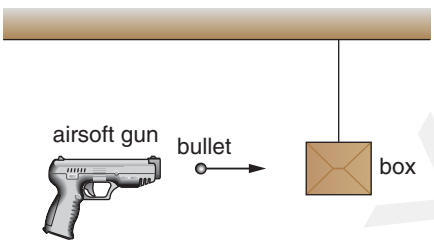


Fig ab

(a) Find the common speed of the bullet and the box just after the collision. (2 marks)
1.17 m s^{-1}

(b) Find the speed of the bullet just before the collision. (2 marks)

(c) Suppose that the KE of an airsoft gun bullet should not exceed 2 J by law. Does this airsoft gun comply with the law? (2 marks)

(d) Is the total momentum of the bullet and the box conserved in the whole process? Explain briefly. (2 marks)

★**32** Misc A tennis ball of mass 0.06 kg is placed on a basketball of mass 0.6 kg (Fig ac). They are released from rest at the same time from 1 m above the ground. Assume that all collisions are elastic and the sizes of the balls are negligible.

Fig ac

(a) Find the velocity of the basketball just before reaching the ground. (2 marks)

(b) The impact between the basketball and the ground takes 0.15 s. Find the average impact force acting on the basketball by the ground. (3 marks)

(c) The basketball collides with the tennis ball just after rebounding from the ground. Assume that the velocity of the tennis ball just before the collision is the same as the answer to (a) and the total momentum of the balls is conserved during the collision between the two balls. State how the KE of each ball changes in the collision. (2 marks)

★**33** Misc A student verifies the law of conservation of momentum using the set-up in Figure ad. A card is attached to each glider. A light-gate can measure the time that an object takes to pass through it. Describe the procedures and the expected result of the experiment. (4 marks)

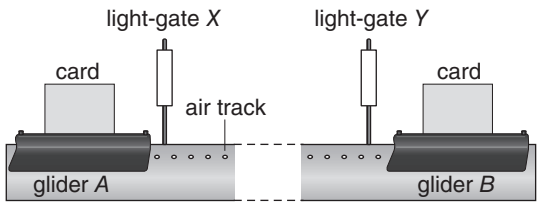


Fig ad

★**34** 7.2 A trolley of mass 0.5 kg is released on a frictionless track which is inclined at an angle of 10° to the horizontal (Fig ae). It collides with a fixed barrier at the lower end of the track. Figure af shows how the force F acting on the trolley by the barrier changes with time t. Figure ag shows how the trolley's velocity v change with t. The direction up the track is taken as positive in both graphs.

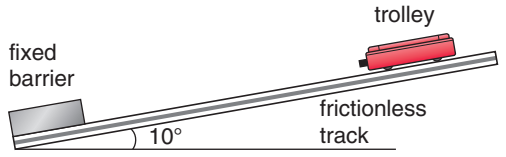


Fig ae

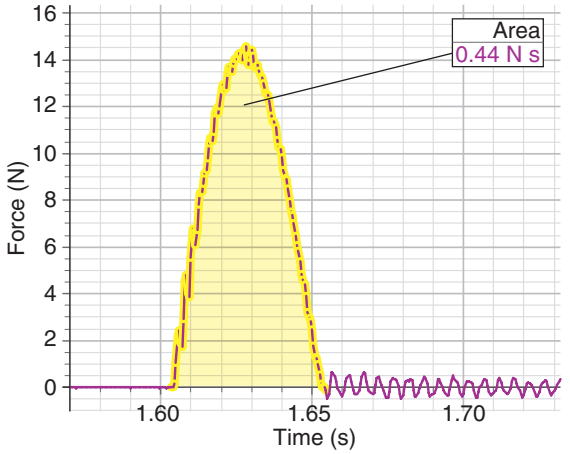


Fig af

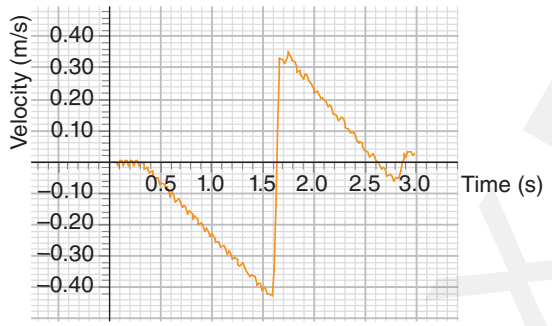


Fig ag

(a) Draw the free-body diagram for the trolley during the impact. (2 marks)

(b) From Figure ag, estimate the change in momentum of the trolley during the impact. (2 marks)

(c) Find the difference between the answer to (b) and the area under the graph in Figure af. Explain the difference. (2 marks)

★★35 7.2 A water wheel turns when a stream of water hits the paddles (Fig ah).

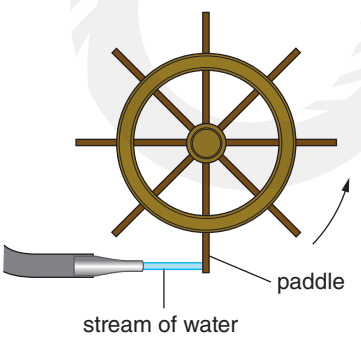


Fig ah

(a) A force acts on the paddle when it is hit by a stream of water. Explain this by considering the momentum of the water. (3 marks)

(b) If curved paddles are used, water can leave the paddle in the opposite direction (Fig ai).

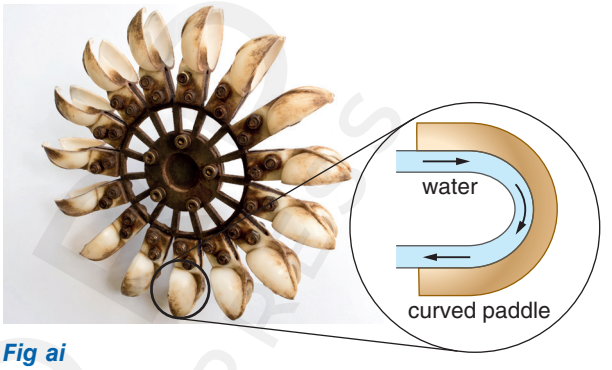


Fig ai

(i) Explain why the curved paddle results in a larger force when it is hit by the same stream of water. (1 mark)

(ii) Suppose water flows at a rate of 5 kg s^{-1} into a fixed curved paddle. The water enters the paddle at 10 m s^{-1} towards the right and leaves at the same speed towards the left. Find the force acting on the paddle by the water. (2 marks)

▶ Refer p.314

★★36 7.1 Carts *A* and *B* are put in contact with each other on a smooth horizontal plane. The mass of *A* is 0.5 kg and that of *B* is 1.5 kg. When the spring-loaded plunger of *A* is released, the two carts move in opposite directions (Fig aj). The elastic potential energy stored in the spring is 5 J and is converted completely into the KE of the carts during the 'explosion'.

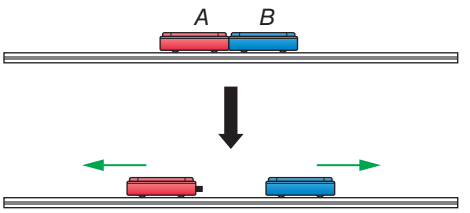


Fig aj

(a) Find the speeds of *A* and *B* after the explosion. (4 marks)

(b) Practically, not all the elastic potential energy is converted into KE during an explosion. Suggest what the remaining elastic potential energy is converted to. (1 mark)

▶ Refer p.291

★★37 Misc

Blocks A and B move along a straight line on a smooth horizontal plane (Fig ak) and collide head-on. The mass of A is 0.4 kg and the mass of B is 0.8 kg.

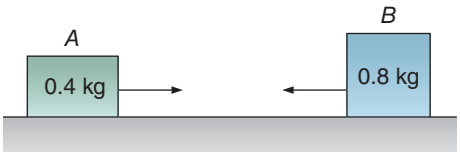


Fig ak

Figure al shows the velocity–time graph of B before and after the collision.

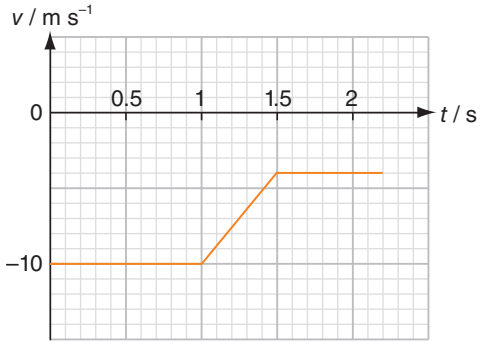


Fig al

(a) Find the net force acting on A during the collision. (3 marks)

(b) A moves at 2 m s^{-1} towards the right at $t = 0$. Draw the v–t graph of A in Figure al. (2 marks)

(c) Is the collision elastic? (1 mark)

Refer p.309

38 Misc

HKDSE 2013 Paper 1B Q6

See p.280 for the Exam Report.

Read the following passage about '**Bungee jumping**' and answer the questions that follow.

Bungee jumping is an activity that involves jumping from a tall structure while the person is connected to it via a thick elastic cord. When the bungee jumper jumps, the cord stretches after falling a certain distance. The bungee jumper is momentarily at rest at the lowest point but then bounces back up into the air. The bungee jumper continues to oscillate up and down a few times before he comes to a complete stop.

Fig am

A simple 'ankle attachment' (as shown in the above photo) can be used to secure the player to the cord. However, due to accidents where the ankle attachment became detached from the bungee jumper, many operators now use a 'full body harness'.

Fig an

When answering the following questions, neglect the effects of air resistance.

(a) (i) Describe the acceleration of the bungee jumper during the first downward fall to the lowest point. (3 marks)

(ii) State the energy change during the period from the beginning of the jump to the moment when the bungee jumper is at the lowest point of his first downward fall. (2 marks)

(b) In terms of the net force acting on the bungee jumper, explain why the cord has to be elastic. (2 marks)

(c) In terms of contact area, explain why a 'full body harness' is less likely to cause injuries to or detach from the bungee jumper than a simple 'ankle attachment' during a fall. (2 marks)

39 Misc **HKDSE 2016 Paper 1B Q4** See p.306 for the Exam Report.

A person wears a water jetpack which enables him to stay 'afloat' in equilibrium in the air as shown in Figure ao. A pump on the sea surface continuously pumps water to the jetpack via a hose and the water is then ejected downwards.

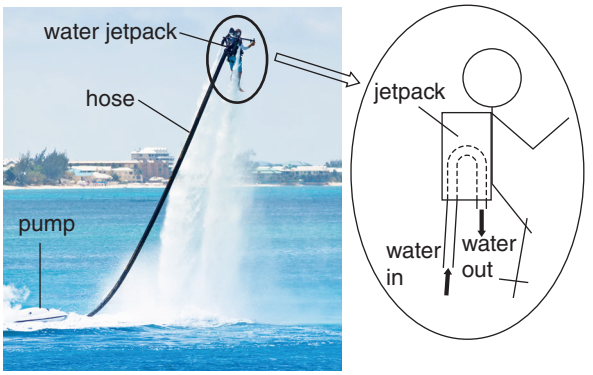


Fig ao

(a) Referring to Figure ao, water enters the U-shape hose inside the jetpack with a certain speed and is then ejected out vertically downwards. Use Newton's law(s) of motion to explain why a lifting force acting on the person is produced. (3 marks)

(b) Draw and label all the forces acting on the person wearing the jetpack as a whole in the free-body diagram below. Neglect the pulling force due to the hose connected to the jetpack. (1 mark)

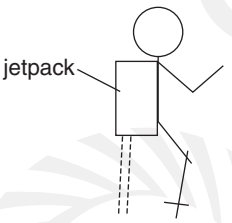


Fig ap

(c) Suppose that water enters the jetpack with a speed of 10 m s^{-1} vertically upwards and is then ejected out at the same speed vertically downwards. (g = 9.81 m s^{-2})

(i) **Just by considering the change of momentum of the water**, estimate how much water, in kg, has to be ejected per second to provide a lifting force of 1000 N needed. (2 marks)

(ii) Water is pumped to the water jetpack at a height of 7.5 m above sea surface and then ejected from it. By considering the gain in mechanical energy of the water, estimate the minimum output power of the pump. (3 marks)

(d) The person changes to staying 'afloat' in equilibrium at a higher position. If the speed by which water enters and is ejected from the jetpack remains the same, would the amount of water ejected per second be greater than, equal to or smaller than the result found in (c)(i)? Explain. (Neglect the weight of the hose.) (2 marks)

40 7.1 **HKDSE 2017 Paper 1B Q2**

The following experimental items are provided to set up an experiment to estimate the speed of a bullet fired from an air gun.

a smooth track

a trolley

a motion sensor used to measure the speed of the trolley

some plasticine

an air gun and bullets

an electronic balance

The set-up is shown in Figure aq.

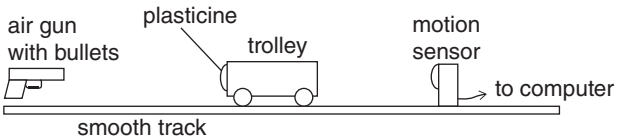


Fig aq

Describe the procedures of the experiment. State the physical quantities to be measured and an equation for finding the speed of the bullet. Write down **one** precaution for getting a more accurate result. (5 marks)

Q40 Exam Report: This question required candidates to describe how to measure the speed of a bullet using the apparatus provided. The general performance was poor. Many candidates failed to mention that the speed of the trolley immediately after the collision should have been taken. A small number of candidates did not know that the motion sensor registered the trolley's speed instead of its distance travelled. Not many were able to state the precautions for getting a more accurate result such as the bullet should be fired along the trolley's direction of travel.